

Aim

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LK1 pilot run

Code ▾

260423_VH01624_453_222HWMYX

2026-08-03

Run: 260423_VH01624_453_222HWMYX

Status: superseded - LK1 was re-sequenced as 260522_VH01624_461_222JLJVN . Everything below is the first pass on the pilot data and should not be quoted as a current result.

Analysis finished: 2026-05-20 - the most recent of the 34 figures shown below.

Report generated: 2026-08-03 by 19.LK1_pilot_run_report.Rmd , from the outputs of steps 01-18 in this folder.

This is the first run of the project and most of the pipeline was worked out on it. Several of the scripts that produced these outputs have since been renumbered or replaced; the copies in this folder are the versions that ran, renumbered into execution order.

Aim

Establish the single-cell workflow for the AML surface-proteome project on the first multiplexed run, before committing to it for the rest of the cohort. Four samples were hashed into a single GEM well with paired GEX, ADT (TotalSeq-A) and HTO libraries, which sets up four questions this run was meant to answer:

- does HTO hashing recover the four samples, and does genotype demultiplexing agree with it
- which ADT normalisation to carry forward, CLR or DSB
- do the cells annotate sensibly against a bone marrow reference
- can malignant cells be called from CNVs in data of this depth

Conclusions

- overall quality of the run was of ok quality, with a potential issue of only 44% saturation level, as reported by Cellranger that will be addressed by resequencing
- genotyping with cellSNP/vireo outperforms HTO hashing
- DSB normalisation of CITE-seq outperforms CLR approach
- cell annotation with Bone Marrow Map works and should be used in the future
- can malignant cells be called from CNVs in data of this depth, but alternative callers should be tested and verified

1. Run, libraries and samples

R1 read counts per library (01.count_reads.sh)

fastq	reads	library
LK1-GEX_S5_R1_001.fastq.gz	475,071,355	GEX
LK1-TotalSeqA-ADT_S6_R1_001.fastq.gz	348,013,031	ADT (TotalSeq-A)
LK1-TotalSeqA-HTO_S7_R1_001.fastq.gz	15,864,696	HTO (TotalSeq-A)

Read counts are line count / 4 on R1. The HTO library is far shallower than ADT, which is the expected ratio but worth remembering when reading the demux figures.

Four samples were hashed into the well. The map is hardcoded in 02.initial_QC_and-HTO_demux.R :

HTO barcode	Sample
HT01-GTCAACTCTTTAGCG	MOLM13 (AML cell line)

HTO barcode	Sample
HT02–TGATGGCCTATTGGG	HBDN206-MNpCT
HT03–TTCCGCCTCTCTTTG	HBDN392-AML-MDS
HT04–AGTAAGTTCAGCGTA	HBDN501-AML-KMT2A

Note the naming drift: the HTO map writes them with underscores (HBDN206_MNpCT), the figures and later metadata use hyphens (HBDN206–MNpCT). Same samples.

HTO demultiplexing outcome

HTO_classification.global	hash.ID	n_cells
Doublet	Doublet	5813
Negative	Negative	3455
Singlet	HBDN392-AML-MDS	3281
Singlet	HBDN206-MNpCT	6813
Singlet	HBDN501-AML-KMT2A	1731
Singlet	MOLM13	1996

Per-sample QC medians

hash.ID	HTO_classification.global	n_cells	median_nCount_RNA	median_nFeature_RNA	median_percent_mt	median_percent_ribo	median_nCount_HTO
Doublet	Doublet	5813	4723	2225	2.994497	21.36240	72
HBDN392-AML-MDS	Singlet	3281	3622	1816	1.885681	17.17778	50
HBDN206-MNpCT	Singlet	6813	3928	2013	2.598923	20.77890	42
HBDN501-AML-KMT2A	Singlet	1731	2468	1352	3.079803	19.38263	49
MOLM13	Singlet	1996	20793	5092	3.374237	25.66427	60
Negative	Negative	3455	4179	2017	2.639442	20.50888	8

2. Data locations

Output directories under /scratch/users/nbartonicek/projects/amgen/results/260423_VH01624_453_222HWMYNX

What	Location
Sequencing facility output	/pipeline/Runs/NextSeq/260423_VH01624_453_222HWMYNX/ProjectFolders/Project_Lev-Kats
Working copy, cluster	/scratch/users/nbartonicek/projects/amgen/raw/260423_VH01624_453_222HWMYNX - Sample_LK1–GEX/ , Sample_LK1–TotalSeqA–ADT/ , Sample_LK1–TotalSeqA–HT0/
Reference	Cell Ranger annotation refdata–gex–GRCh38–2024–A , under annotation/

dir	first	last	files
cite_seq_count	2026-04-28	2026-05-05	48
seurat_demux	2026-04-28	2026-05-11	453
emptydrops	2026-05-01	2026-05-01	4
genotype_demux	2026-05-03	2026-05-04	12
scDbfFinder	2026-05-03	2026-05-03	5
demux_comparison	2026-05-04	2026-05-05	32
seurat_annotated	2026-05-05	2026-06-15	195
data_integration	2026-05-06	2026-06-01	517

3. Timeline

What was written, by day

date	directories	files
2026-04-28	cite_seq_count, seurat_demux	27
2026-04-29	seurat_demux	7
2026-05-01	cite_seq_count, emptydrops	22
2026-05-03	cite_seq_count, genotype_demux, scDbIFinder, seurat_demux	47
2026-05-04	cite_seq_count, demux_comparison, genotype_demux, seurat_demux	47
2026-05-05	cite_seq_count, demux_comparison, seurat_annotated, seurat_demux	396
2026-05-06	data_integration, seurat_annotated, seurat_demux	131
2026-05-07	data_integration	12
2026-05-08	data_integration, seurat_annotated	33
2026-05-11	seurat_demux	17
2026-05-12	seurat_annotated	37
2026-05-19	seurat_annotated	1
2026-05-20	data_integration, seurat_annotated	26
2026-05-21	data_integration, seurat_annotated	76
2026-05-25	data_integration	38
2026-05-27	data_integration	207
2026-06-01	data_integration	128
2026-06-09	seurat_annotated	2
2026-06-15	seurat_annotated	12

4. What was run, in order

The scripts in this folder are numbered in execution order and every step of the pilot is accounted for. Steps 02, 04 and 05 were later re-pointed at a subsequent run, so their frozen copies carry the wrong `run` - set it back to the pilot run to reproduce. Each says so in its header. The re-pointing of step 02 was incomplete, which is part of how it was identified: `RUN` and `SAMPLE_SHORT` name LK2 while `SAMPLE` is still LK1-GEX_S5 .

Step	Script	What it does
01	01.count_reads.sh	R1 read counts per library.
02	02.cellranger.sh	Cell Ranger 10.0.0 count on the GEX library.
03	03.initial_QC_and-HT0_demux.R	Seurat object, HTO + ADT attached, CLR, <code>HT0Demux(positive.quantile = 0.80)</code> .
04	04.emptydrops.R	<code>emptyDrops</code> on the raw matrix, $FDR \leq 0.01$. Produces the barcode whitelist.
05	05.cellSNP_vireo.sh	<code>cellsnp-lite</code> pileup over those barcodes, then <code>vireo</code> with <code>N_DONORS=4</code> .
06	06.demux_vireo_and_doublets.R	<code>Vireo</code> calls onto the object, HTO cross-tab, doublet detection.
07	07.CITE_qc.R	CITE QC on <code>emptyDrops</code> cells: raw ADT, CLR, marker distributions, overlays.
08	08.evaluate_DSB.R	DSB using raw background droplets, with IgG-control subsampling.
09	09.combine_cite.R	Folds CITE assays into one object, writes the <code>analysis_bundle</code> .
10	10.annotate_reference_setup.R	BoneMarrowMap Symphony reference, uwot path, first projection.
11	11.annotate_ADT_CLR.R	Projection on the ADT/CLR object.
12	12.annotate_CITE_DSB.R	Same projection on the CITE_DSB object.
13	13.compare_DSB_vs_CLR_one_sample.R	DSB vs CLR in one sample (HBDN206-MNpCT).

Step	Script	What it does
14	14.compare_DSB_vs_CLR_all_samples.R	DSB vs CLR across all samples. Writes data_integration/ 01-06.
15	15.data_integration_composition.R	Composition by sample, DSB protein-space UMAP, CopyKAT against annotation. Writes data_integration/ 07-11.
16	16.copykat_pooled.R	CopyKAT, all cells pooled.
17	17.copykat_per_sample.R	CopyKAT per donor.
18	18.surfaceome_trial.R	First surfaceome pass. Exploratory, saves nothing.
19	19.LK1_pilot_run_report.Rmd	This report.

Key parameters in step03: `CreateSeuratObject(min.cells = 3, min.features = 50)`, `percent.mt` from `^MT-`, HTO and ADT normalised CLR with `margin = 2`, `HTODemux(positive.quantile = 0.80)`, RNA scaled regressing `percent.mt`, PCA and UMAP on `dims = 1:30`, `FindClusters(resolution = 0.5)`.

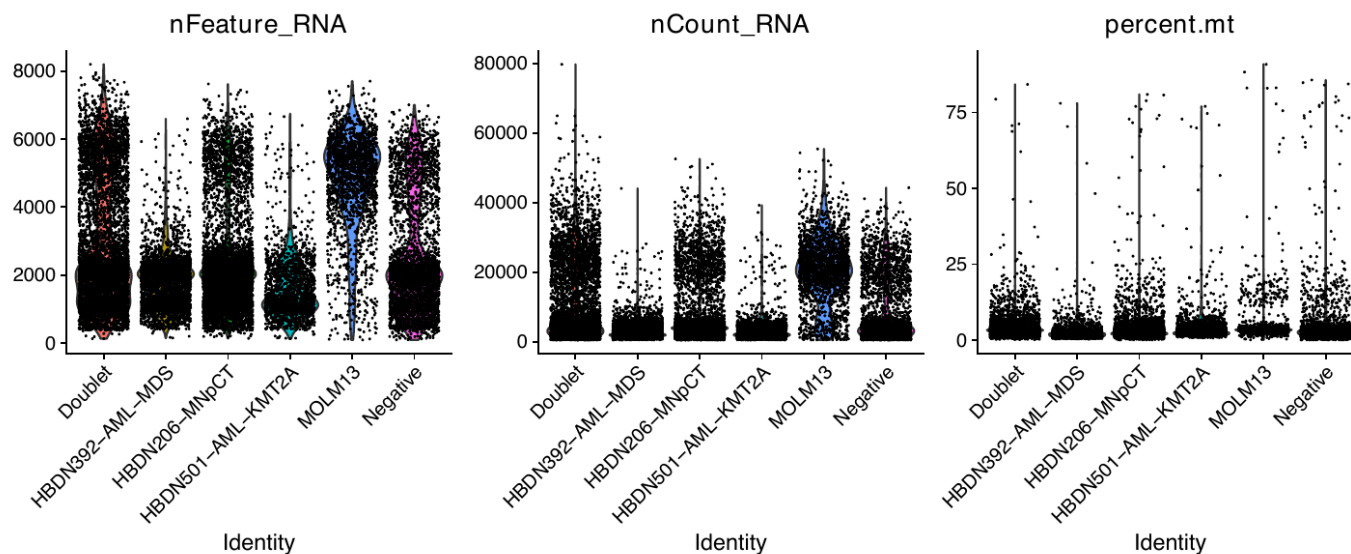
The 0.80 positive quantile is the value that later got swept - `demux_comparison/` carries `_all_positive_quantiles` and `_positive_quantile_0.99` variants of every figure.

5. Results

5.1 First-pass QC and HTO demultiplexing

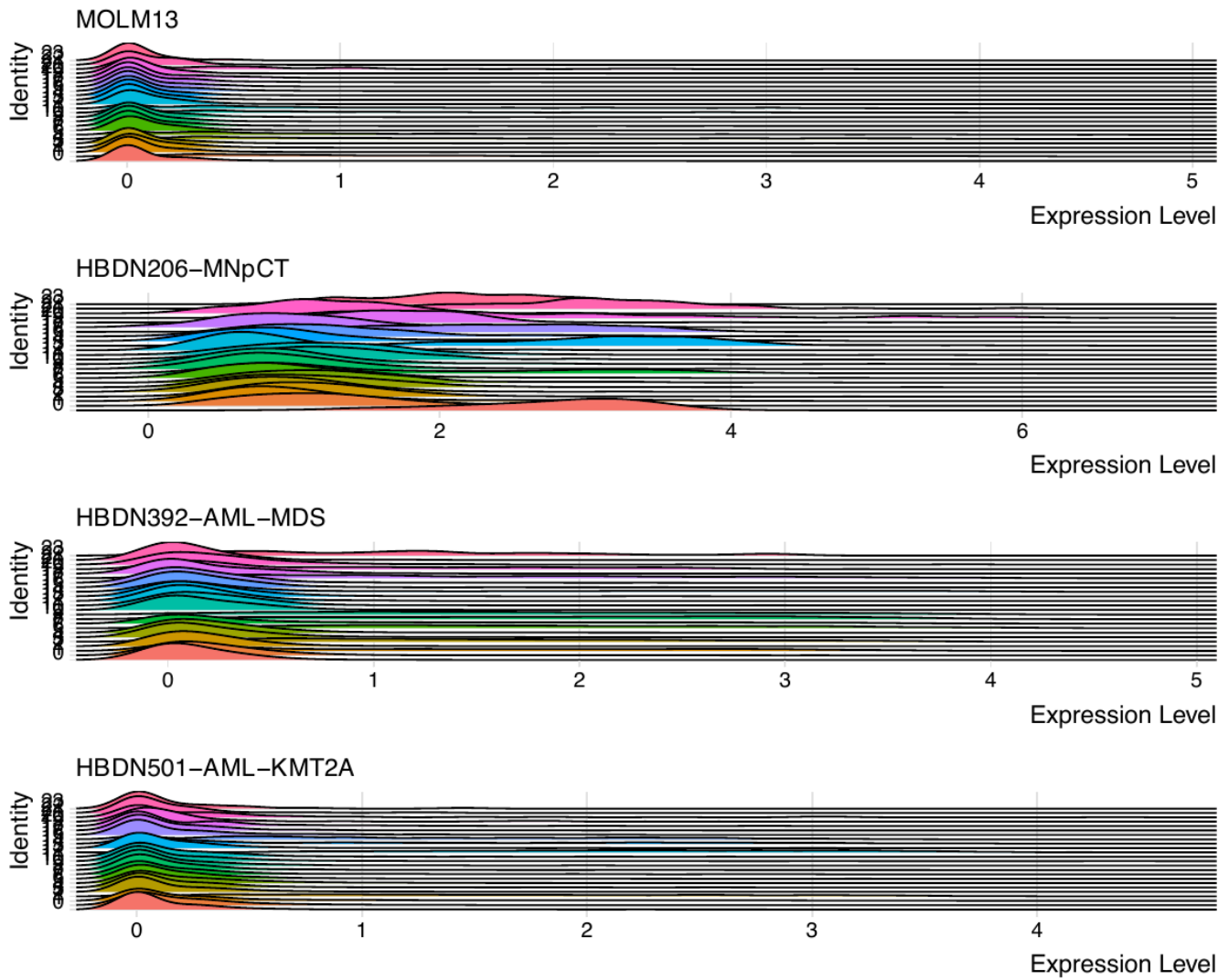
01_initial_qc_violin.pdf

`nFeature_RNA`, `nCount_RNA` and `percent.mt` by `hash.ID`. MOLM13 had the larges amount of RNA and genes detected as expected. No hashed sample was more degraded relative to the others, based on HTO alone.



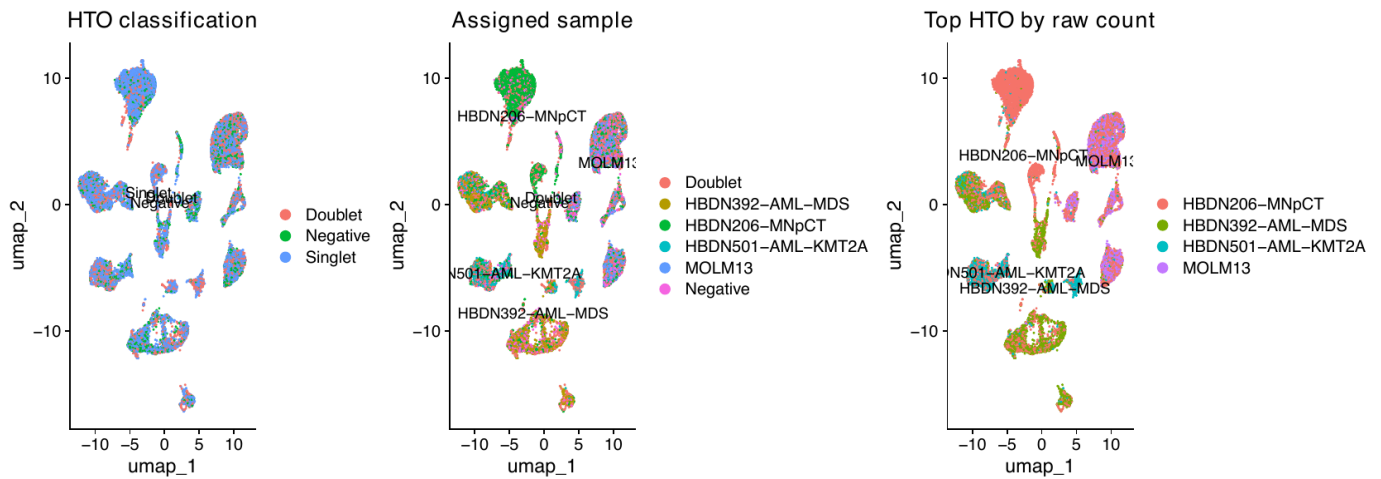
02_hto_ridgeplots.pdf

CLR-normalised HTO signal per hashtag. Lack of clean bimodality per HTO points to issues with HTO tagging.



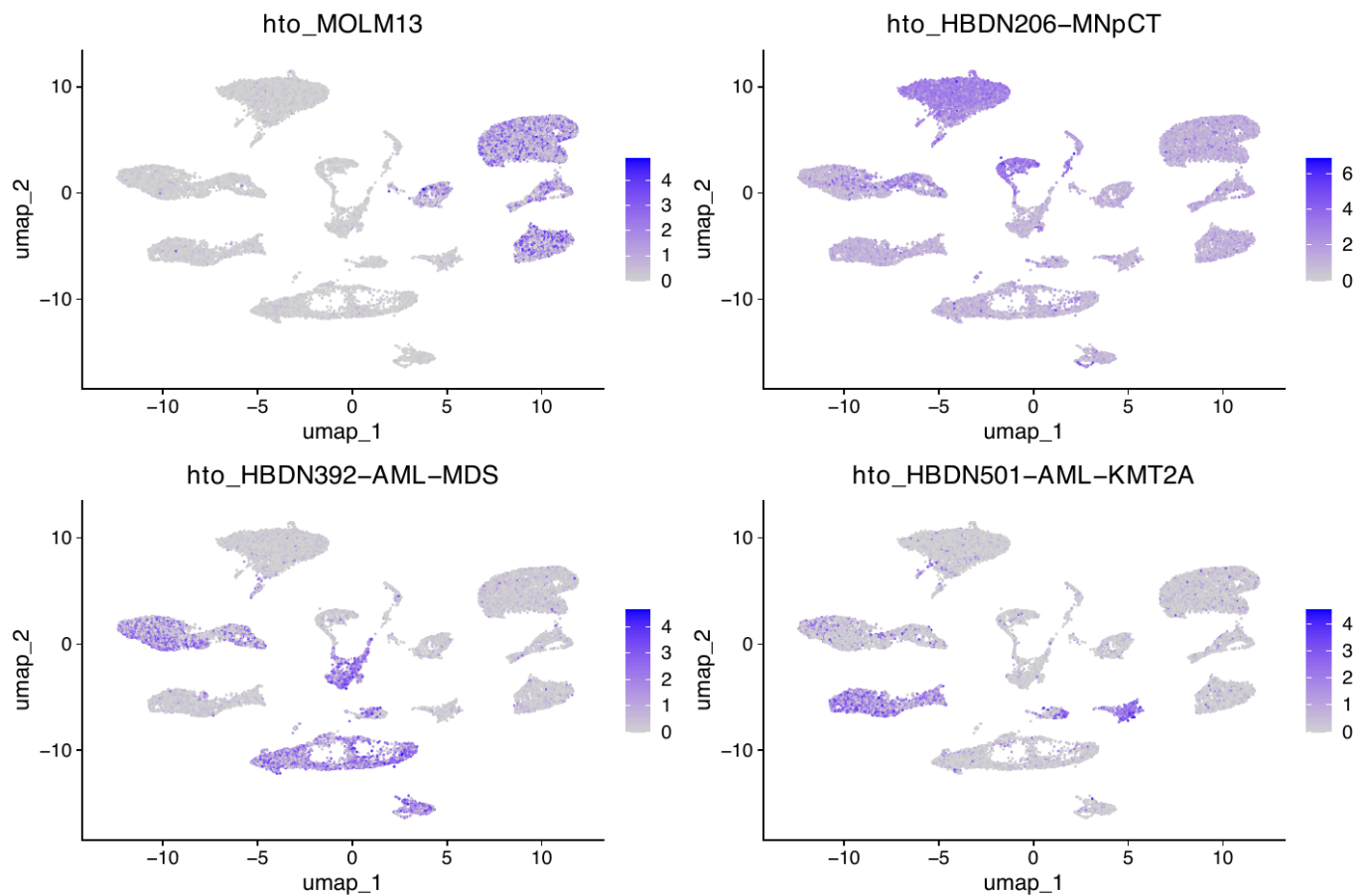
03_umap_demux.pdf

HTO global classification, assigned hash.ID, and top HTO by raw count. Some clusters that cleanly separate by sample have large number of negative detections.



05_hto_featureplots.pdf

Each HTO on the UMAP. Confirms the hashed samples occupy distinct territory rather than smearing, but that the quality is not great.



5.2 Genotype demultiplexing: HTO vs vireo

Since hashing with HTOs was not robust, we attempted demultiplexing with genotype-based tools: cellSNP and vireo. cellSNP’s “donors” were assigned to individual samples based on the largest overlap with HTO-assigned cells. That doubled the number of uniquely demultiplexed cells, and consolidated the usage of this approach for the future.

Demux efficiency, HTO vs vireo

method	total_cells	singlets	doublets	negatives_or_unassigned	singlet_fraction	doublet_fraction	negative_or_unassigned_fraction
HTO	23520	11457	778	11285	0.4871173	0.0330782	0.4798044
cellSNP_vireo	23520	22405	1115	0	0.9525935	0.0474065	0.0000000

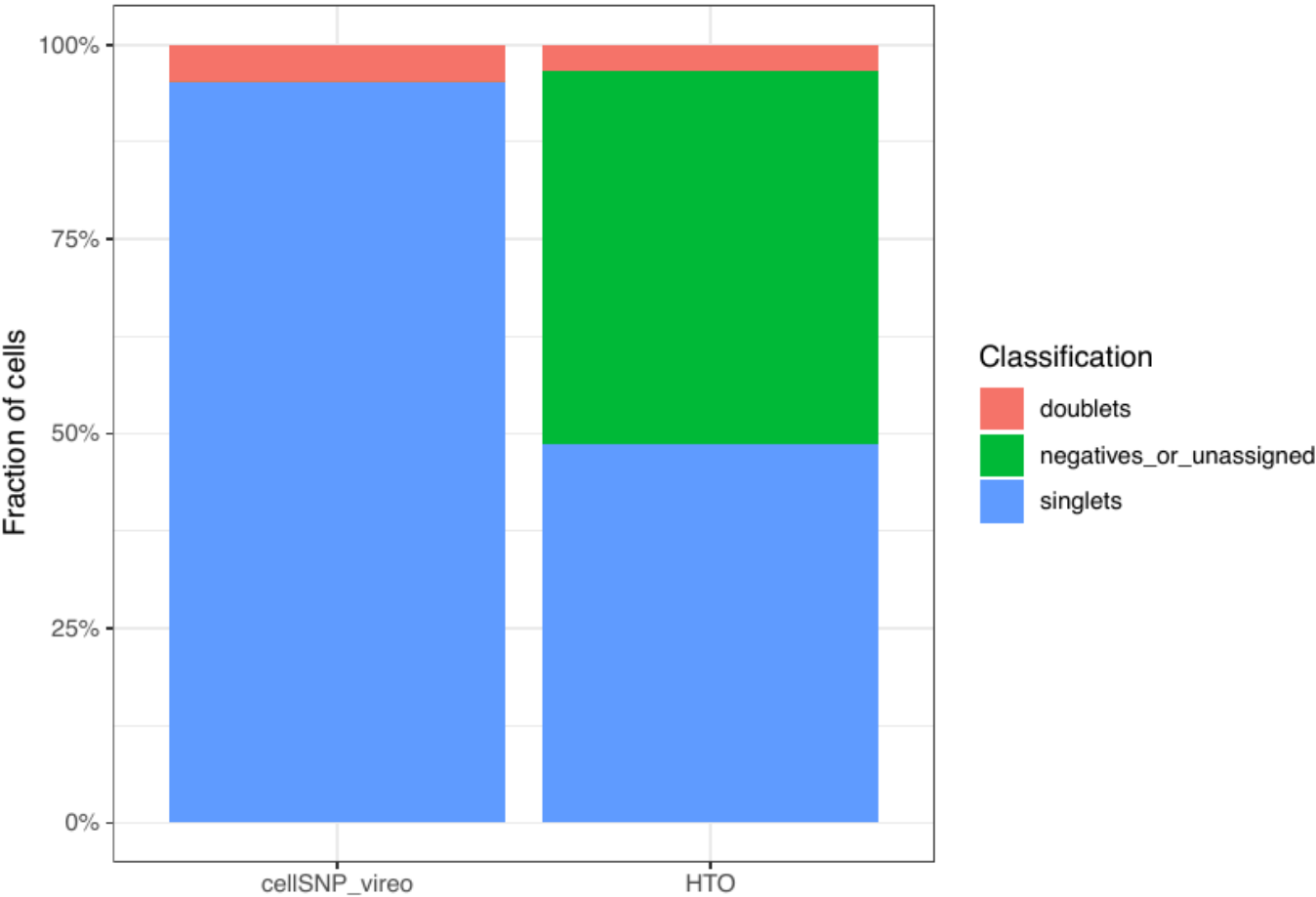
HTO / vireo agreement

total_cells	both_singlet	HTO_doublet_only	Vireo_doublet_only	both_doublet	HTO_singlet_Vireo_unassigned	HTO_negative_Vireo_singlet	both_singlet_frac
23520	10982	538	875	240	0	10885	0.4669

01_demux_efficiency_barplot.pdf

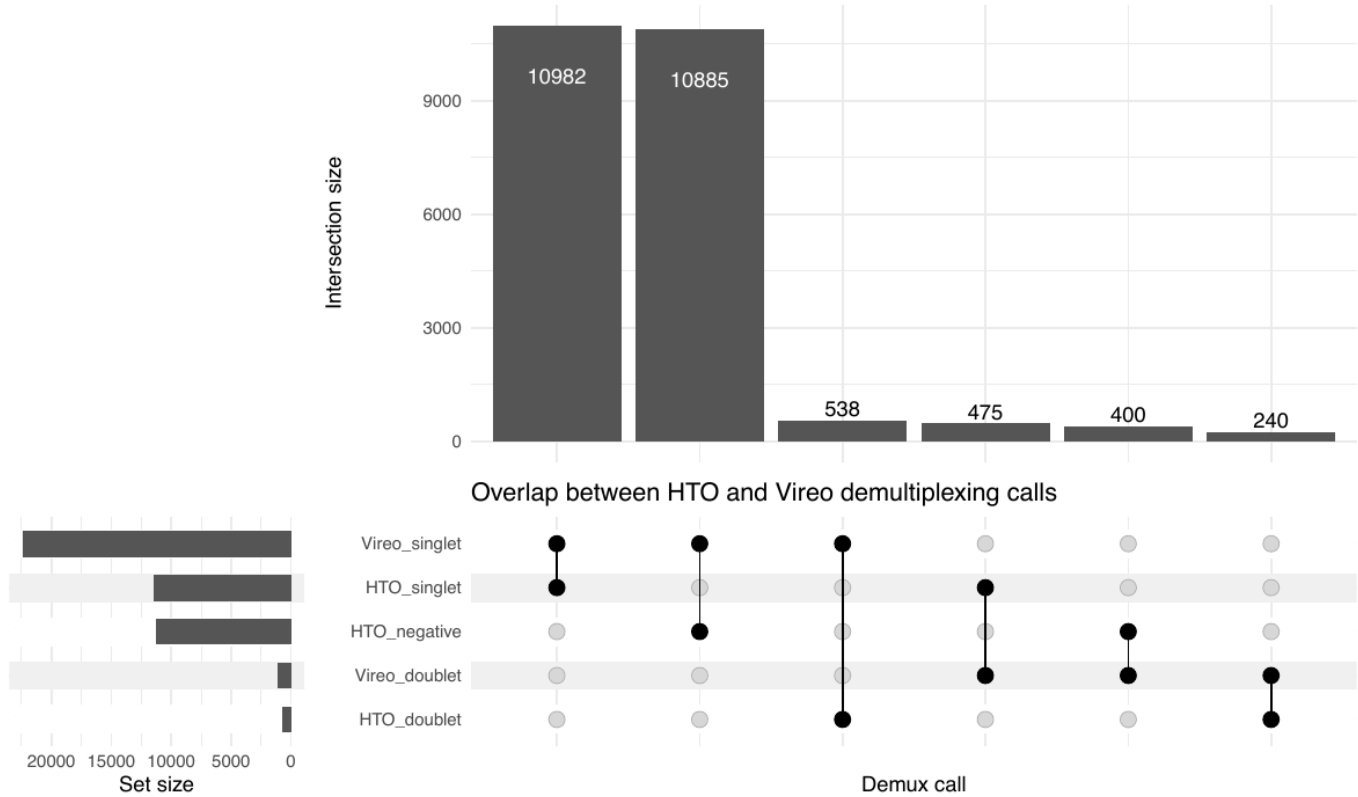
Singlet / doublet / negative rates for HTO and vireo side by side. The headline comparison.

Demultiplexing efficiency: HTO vs cellSNP-vireo

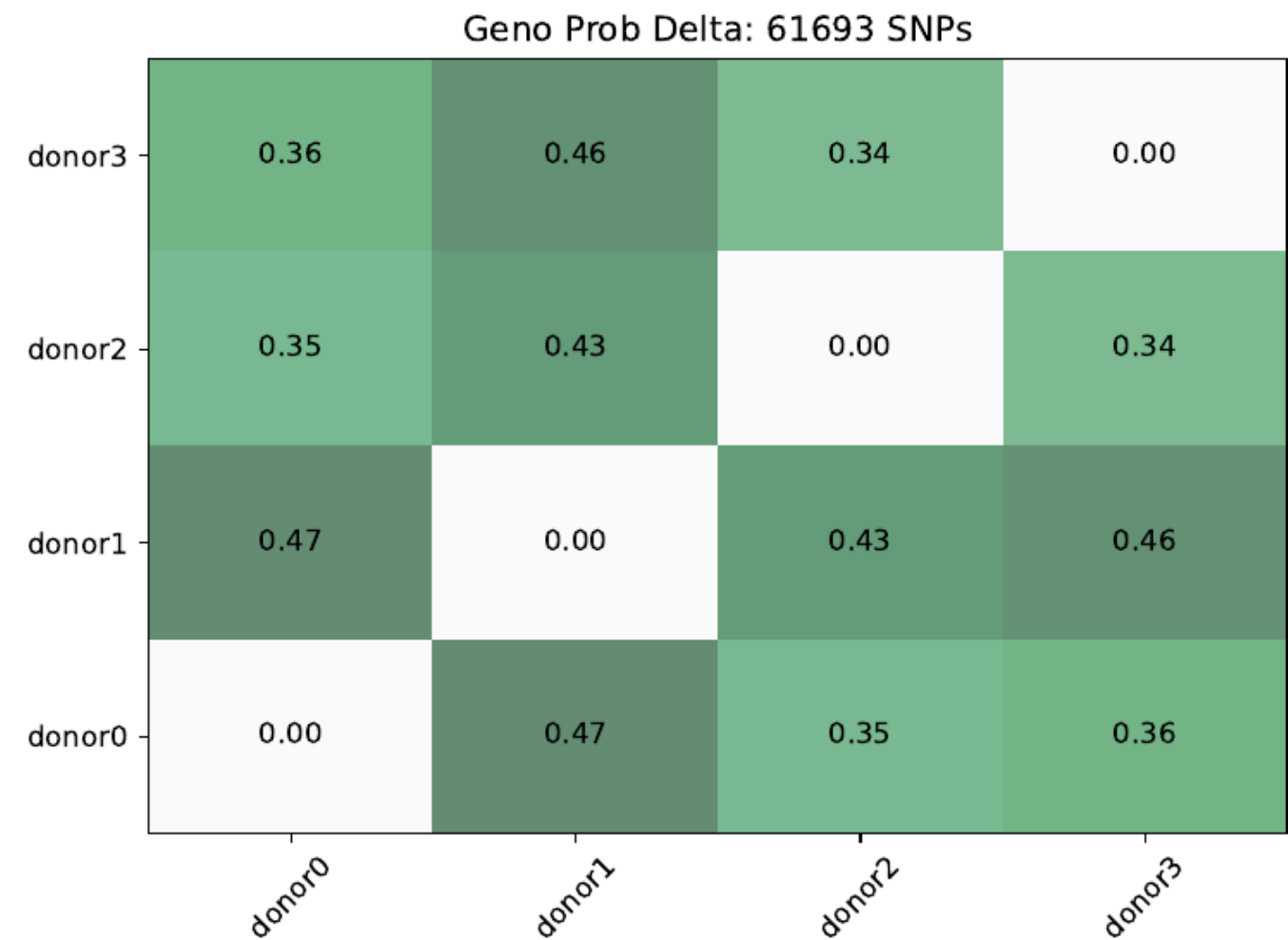


04_upset_HTO_vireo_overlap.pdf

UpSet of the call overlap.



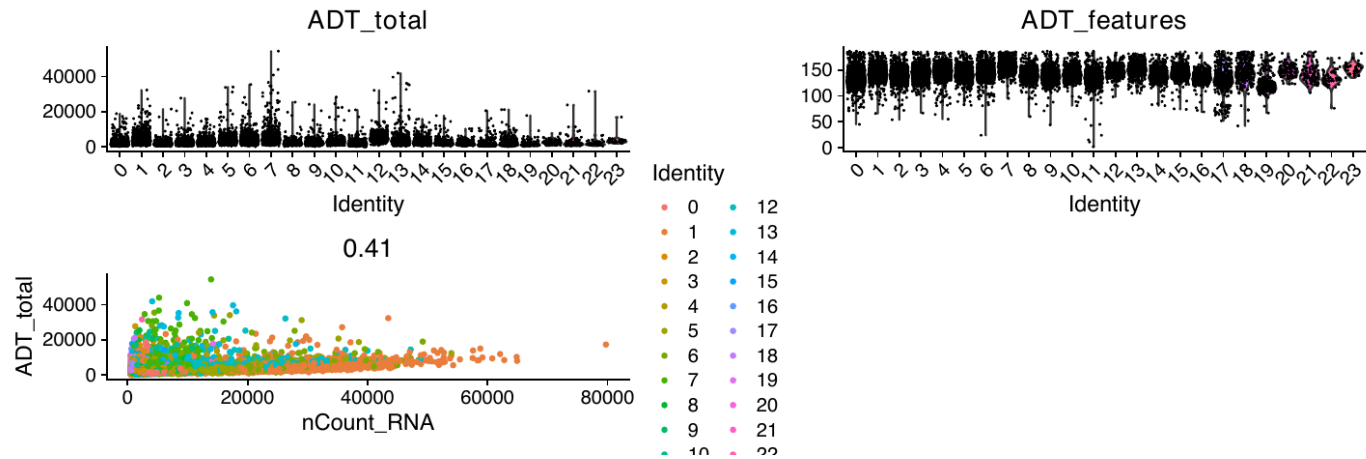
Vireo estimated genotype distance between donors. Confirms the donors are genetically distinguishable in the first place.



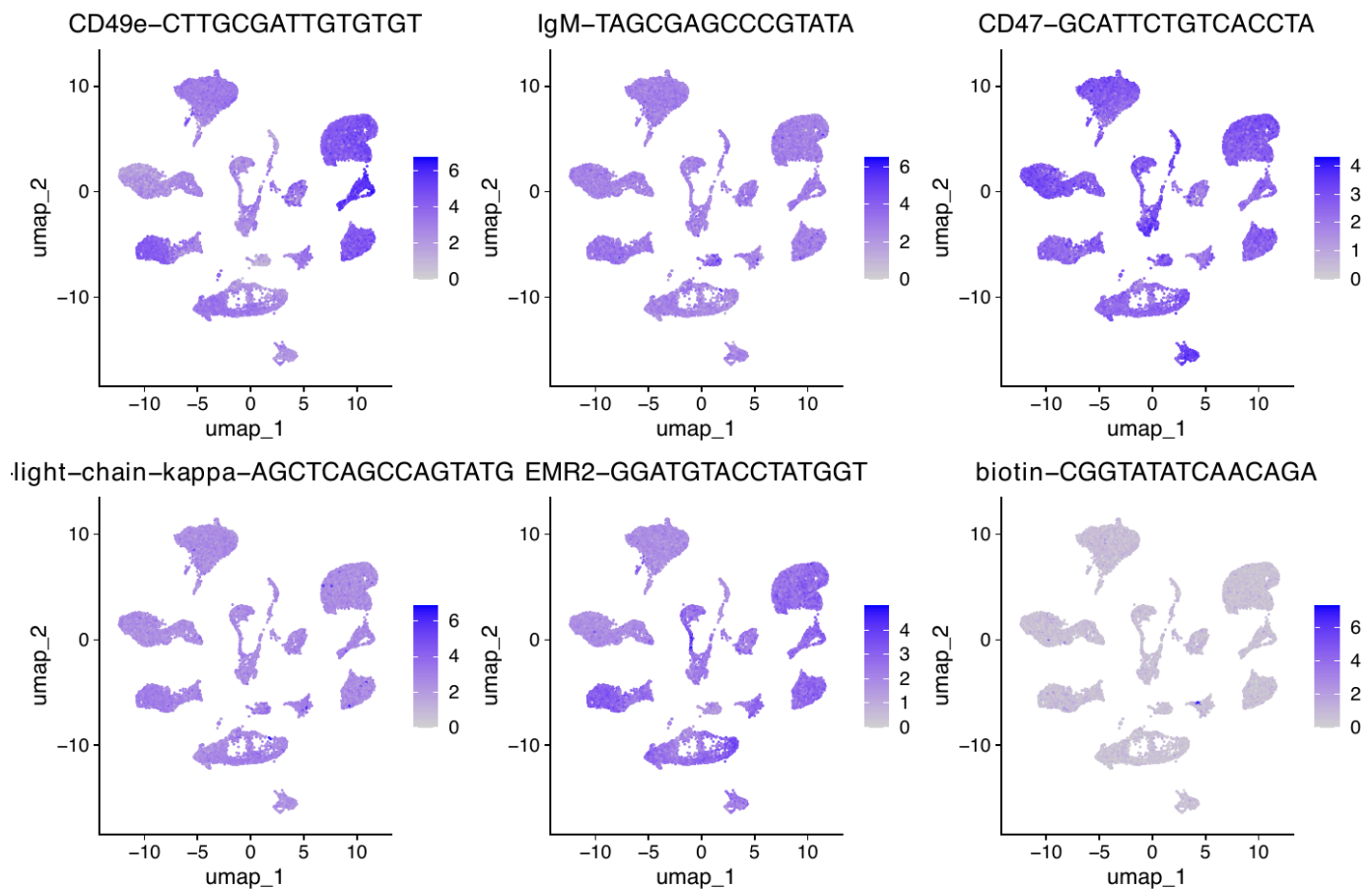
5.3 ADT / CITE-seq QC

CITE-seq performed well, with a number of markers being present only in selected clusters above background. Biotin (as a marker for glycosylated proteins bound by biotinylated I-PHA) was not detectable.

ADT totals and features per cell.

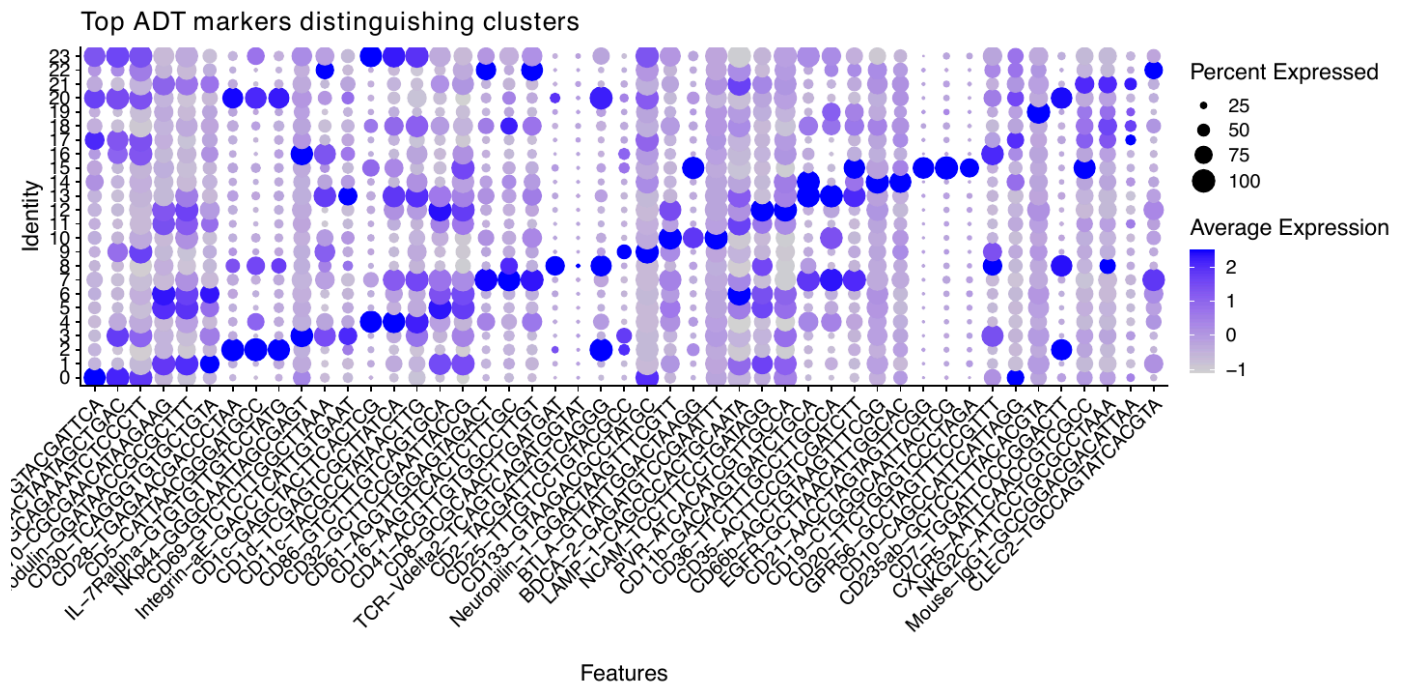


Top 5 ADT markers by mean CLR plus biotin. Biotin was added deliberately here - this is the start of the biotin thread that runs through to July.



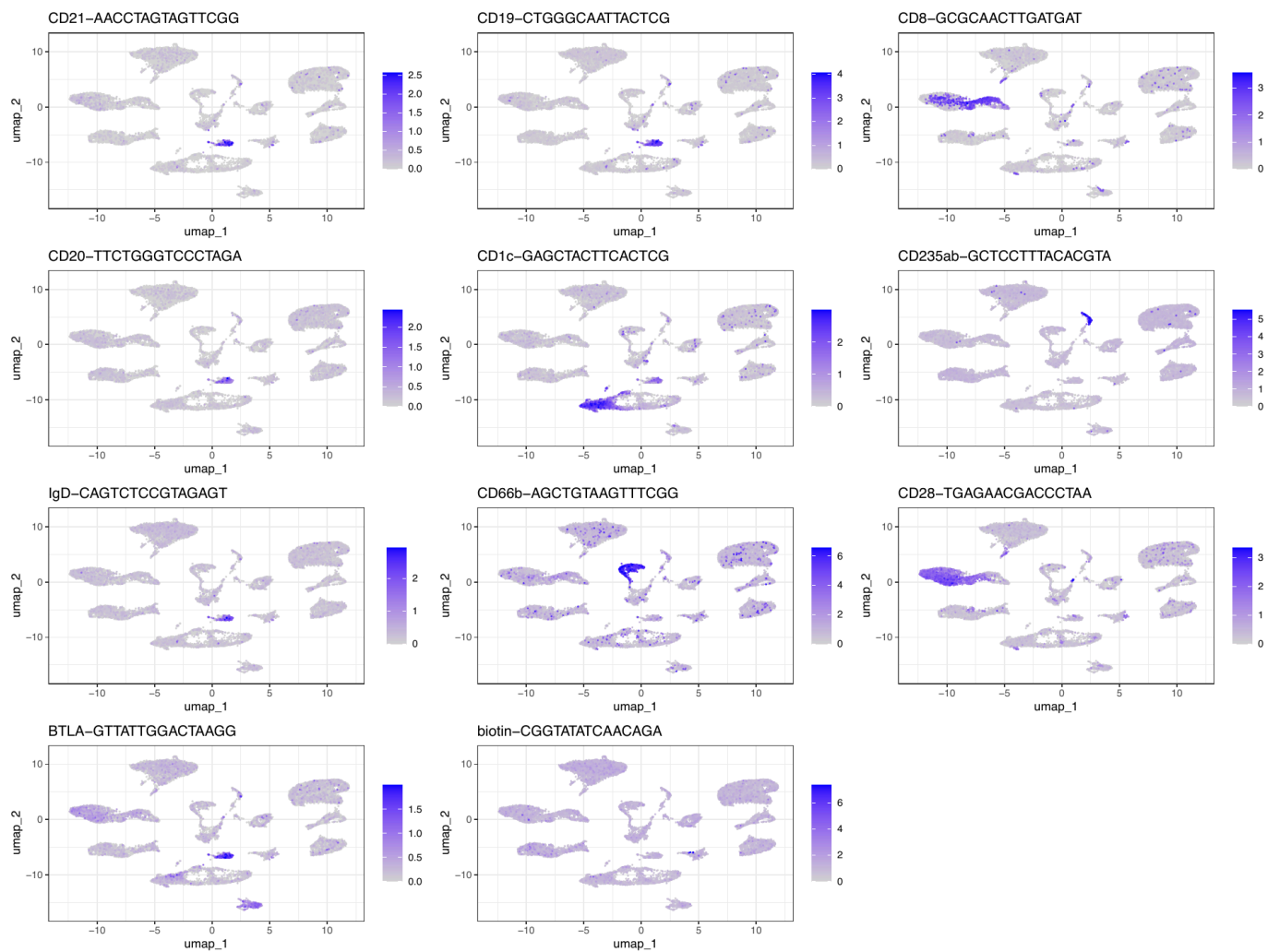
08_top_ADT_cluster_markers_dotplot.pdf

Top ADT markers per RNA cluster.



10_top10_discriminative_ADT_plus_biotin_UMAP.pdf

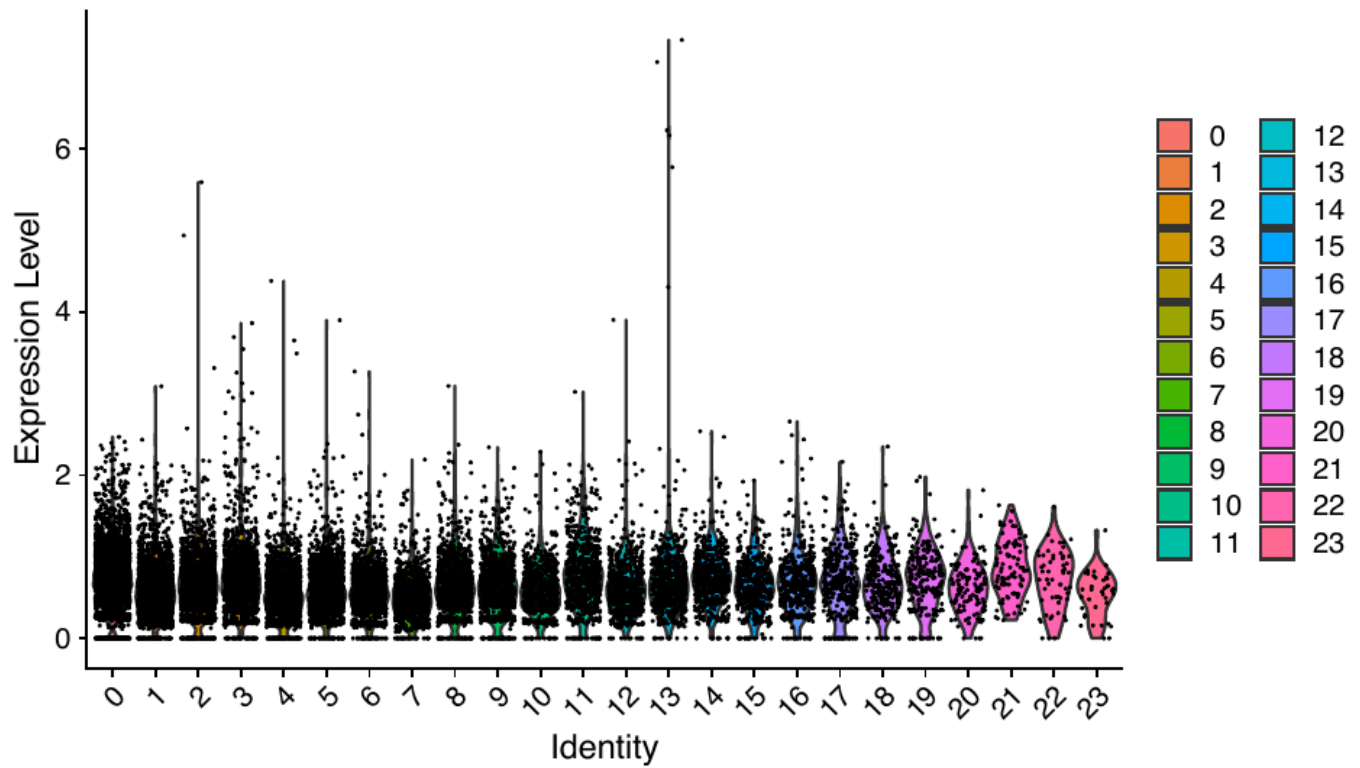
Most discriminative ADTs with biotin overlaid.



11_biotin_expression_by_cluster_violin.pdf

Biotin by cluster - the first look at the heterogeneity later split high vs low.

Biotin expression across clusters



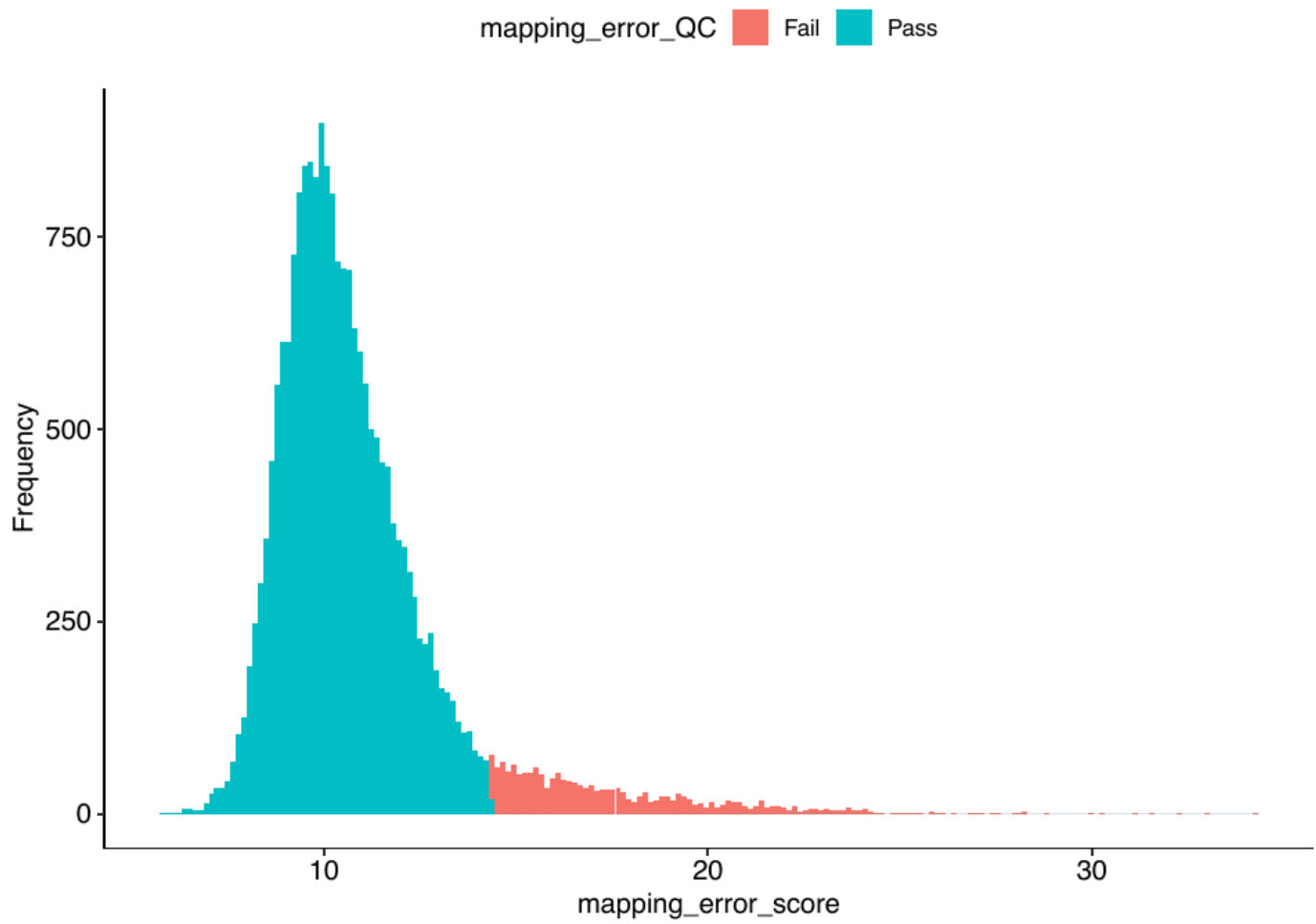
5.4 Annotation (BoneMarrowMap)

Cell annotation was performed by applying tools used in Zeng et al (2025) Bone Marrow Atlas.

01_mapping_error_QC.pdf

BoneMarrowMap mapping error. This is the QC that decides which cells get an annotation at all.

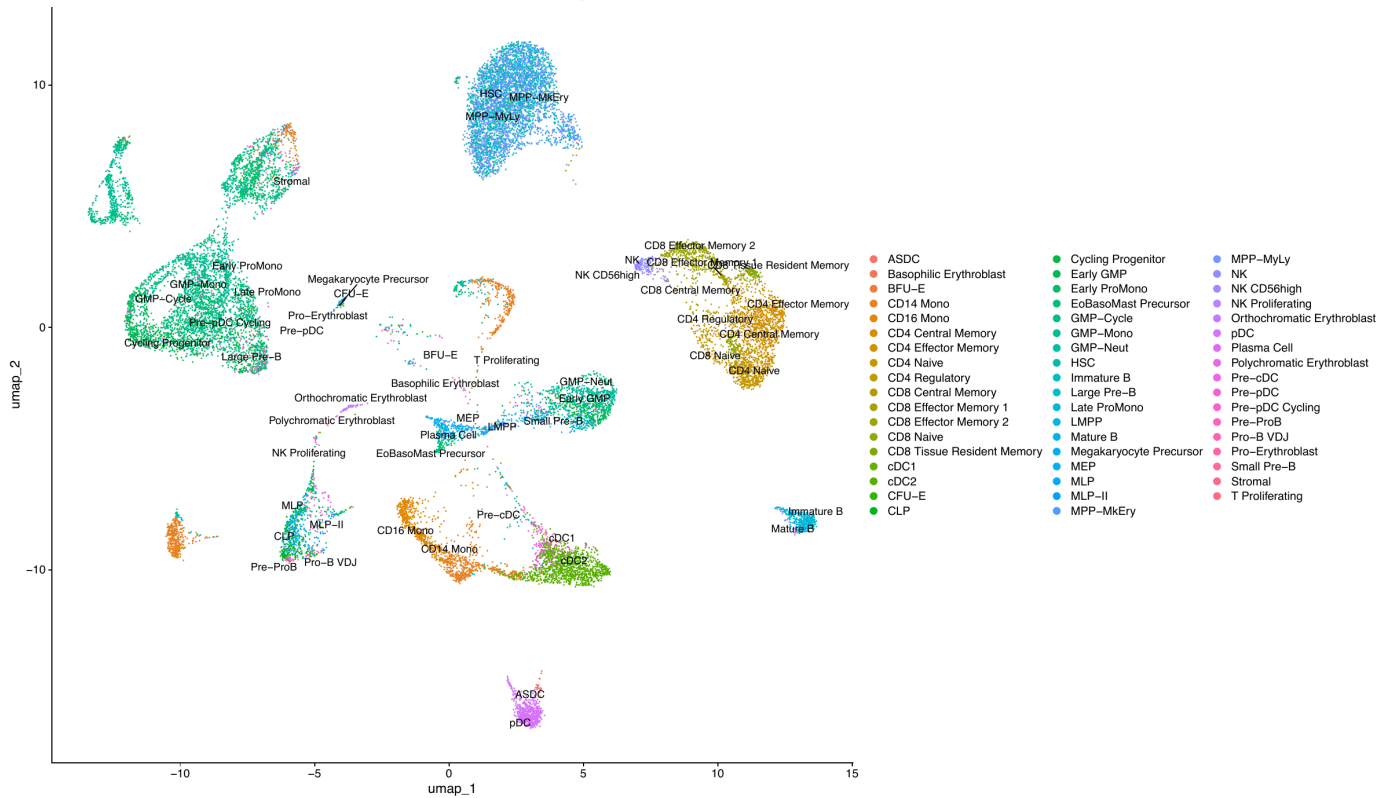
Mapping Error Cutoff



02_projected_predicted_celltypes_pass_only.pdf

Predicted cell types on the reference UMAP, cells passing mapping QC only.

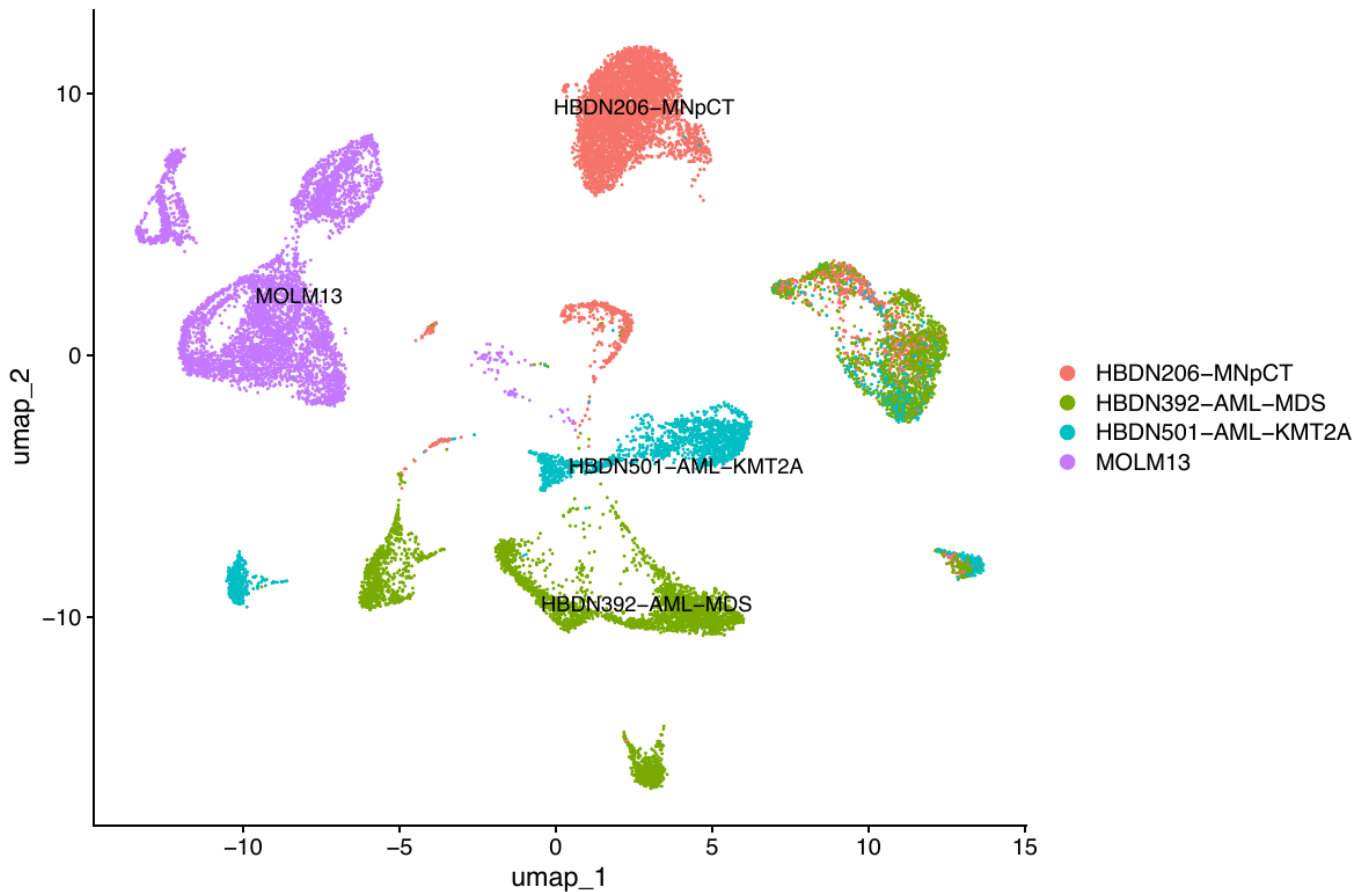
BoneMarrowMap projected cell types: mapping pass only



03_projected_samples_pass_only.pdf

Samples on the projected UMAP.

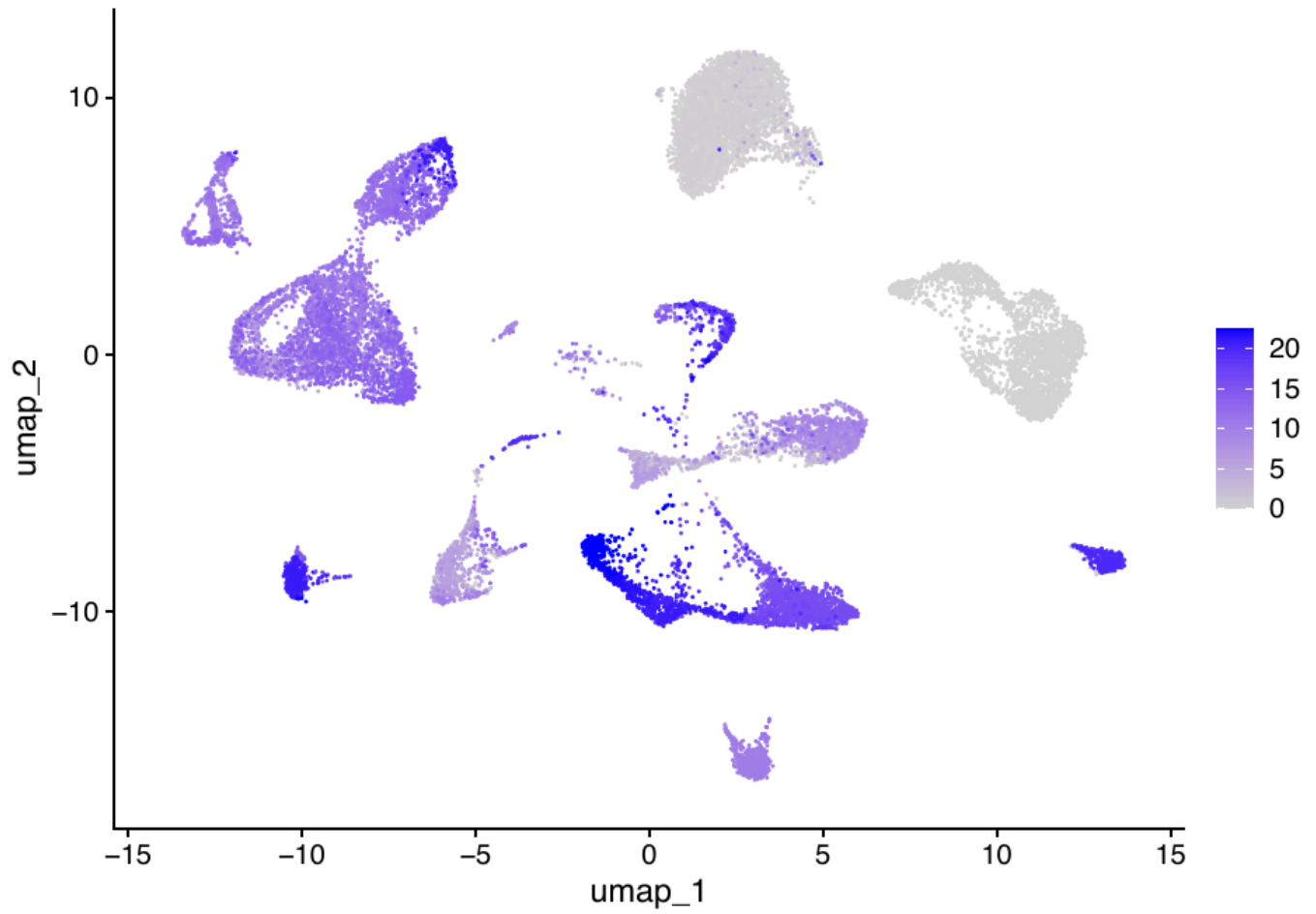
Samples on BoneMarrowMap projection: mapping pass only



04_projected_pseudotime_pass_only.pdf

Projected pseudotime.

Predicted pseudotime: mapping pass only



05_projection_by_sampleID_BMM_helper_manual_split.pdf

Per-sample projection using the BoneMarrowMap helper.

HBDN206-MNpCT

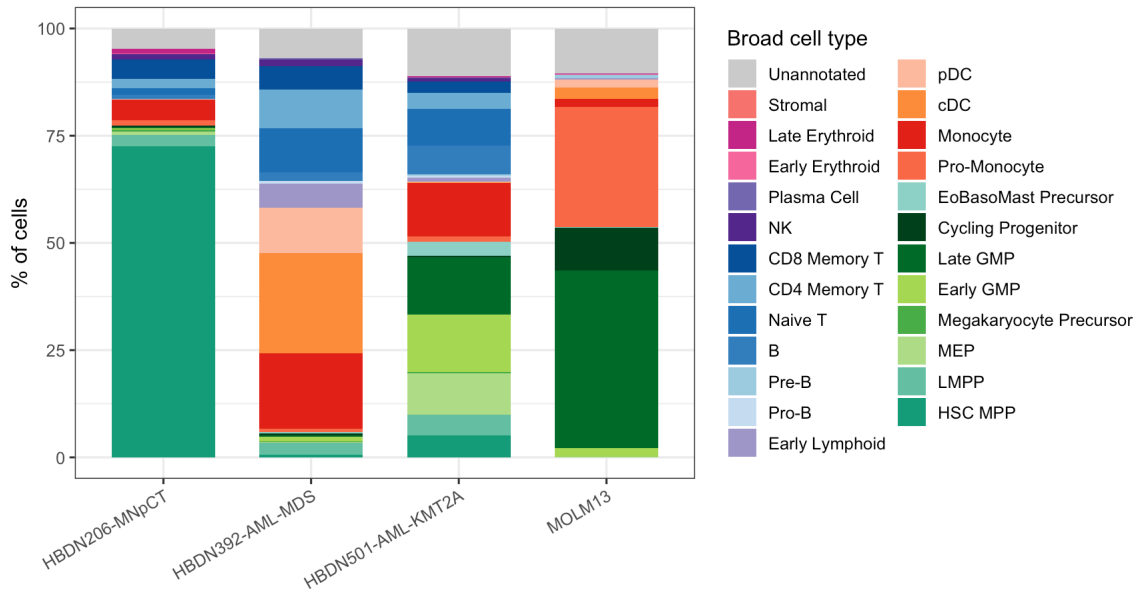
HBDN392-AML-MDS

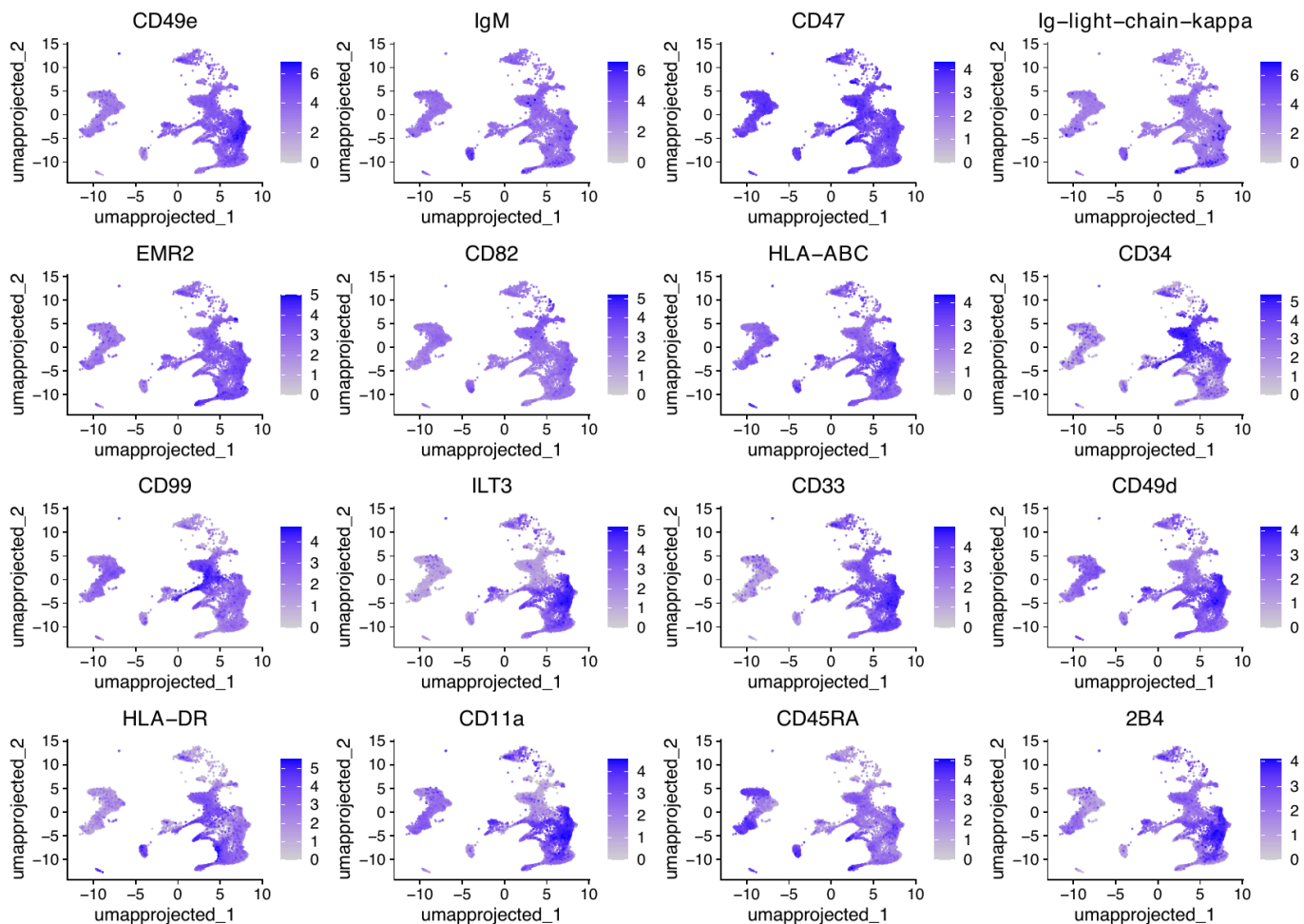
HBDN501-AML-KMT2A

MOLM13

Composition is recomputed here from the exported tables rather than taken from the figures the pilot scripts saved. Those figures keyed the lineage mapping on the *fine* BoneMarrowMap labels (HSC , MPP-MyLy , MPP-MkEry) while the mapping in `annotate.R` expects the *broad* ones (HSC MPP , LMPP), so nearly every cell fell through to "Other" - the saved `10_lineage_composition_by_sampleID.csv` puts 94% of HBDN206-MNpCT there. Reading the broad table instead fixes that, and lets the bars use the pipeline's own palette.

Broad cell type composition per sample

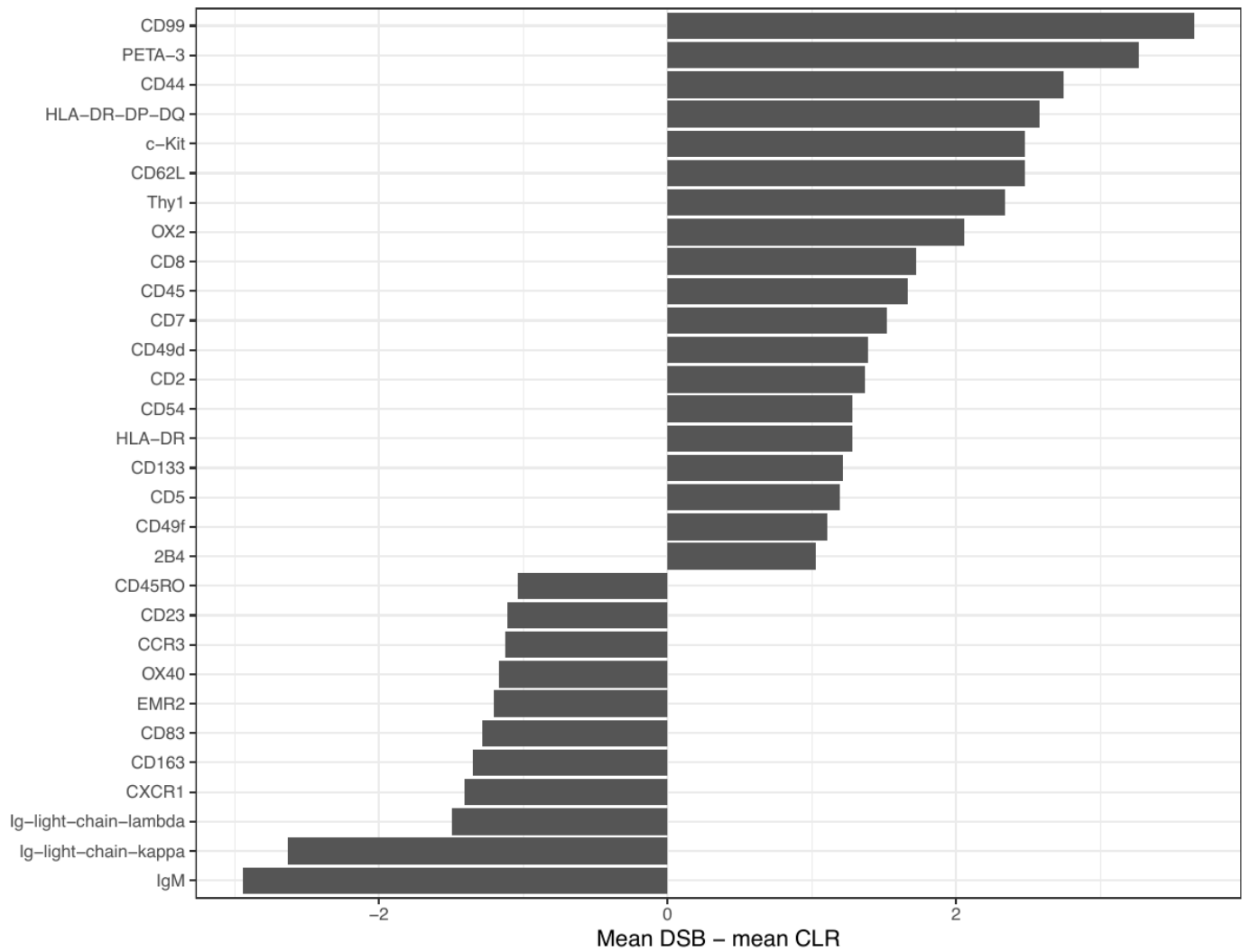




01_marker_mean_shift_DSB_minus_CLR.pdf

Per-marker mean shift, DSB minus CLR, for HBDN206-MNpCT (step 13).

Markers most changed by DSB: HBDN206-MNpCT

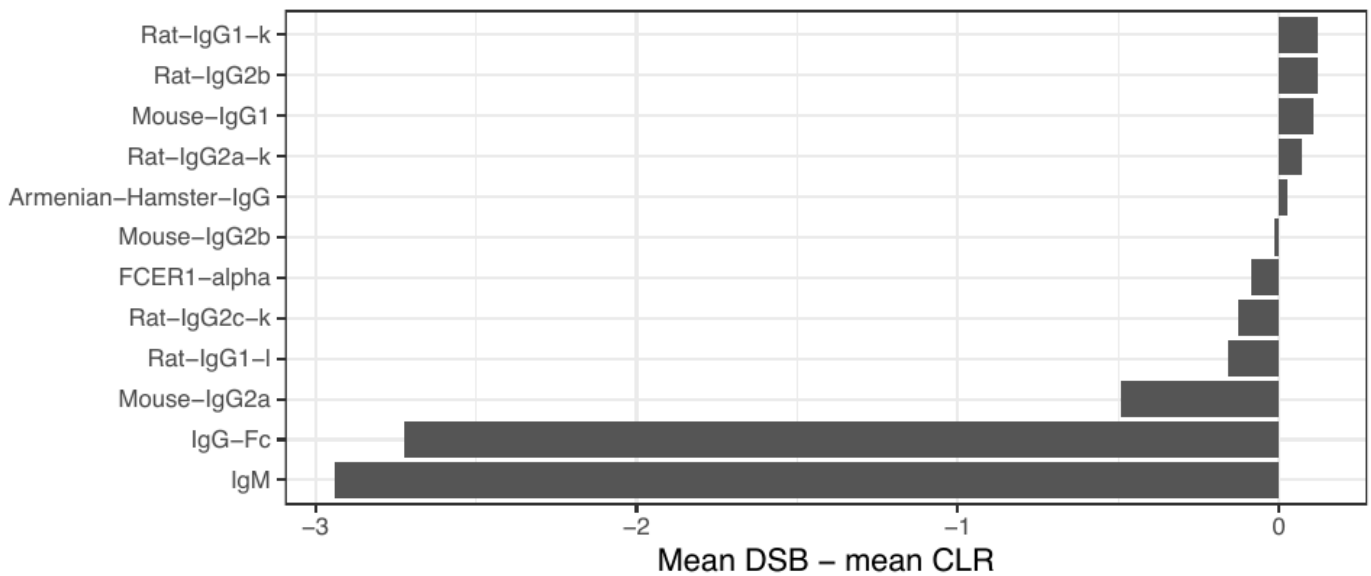


01_background_suppression.pdf

How much background DSB removes, measured on the isotype / background markers. The main argument for using DSB.

Background / isotype suppression

Negative values mean DSB reduced background relative to CLR

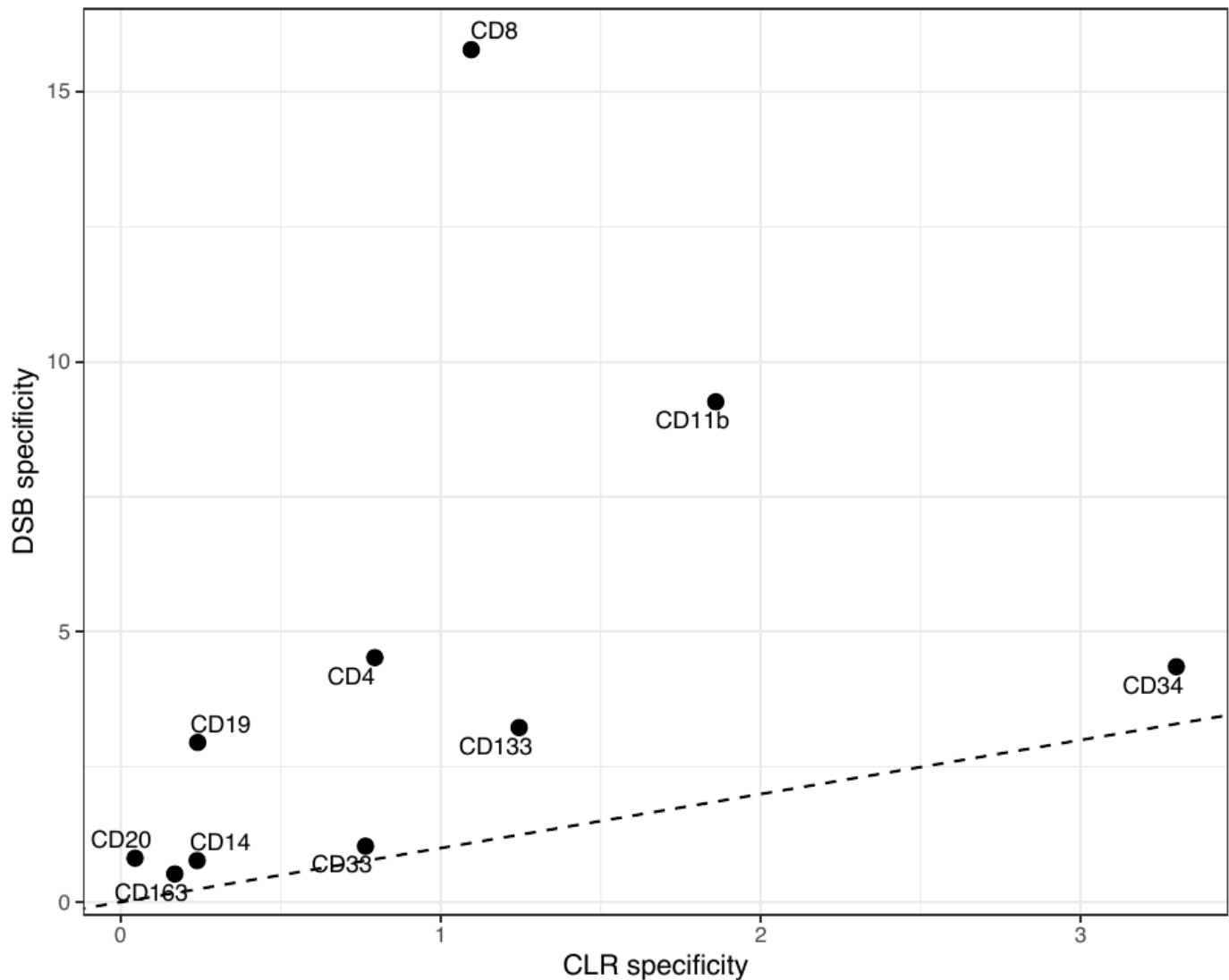


02_expected_marker_specificity.pdf

Whether expected markers land on the expected cell types, scored against a hand-curated panel defined in step 14: CD34 / CD117 / CD133 for stem-progenitor, CD33 / CD14 / CD163 / CD11b for myeloid-DC, CD4 / CD8 / CD19 / CD20 / BCMA / Syndecan-1 for lymphoid, and CD71 for erythroid. For each marker the script takes the **mean signal in its expected broad cell type minus the median across all other broad types**, computes that separately under CLR and under DSB, and plots one against the other with the diagonal drawn. Points above the diagonal are markers DSB made more specific. The underlying numbers, including the per-marker delta, are in `02_expected_marker_specificity.csv`.

Expected marker specificity: DSB vs CLR

Above diagonal means marker is more cell-type-specific after DSB



03_marker_leakage_diffusion.pdf

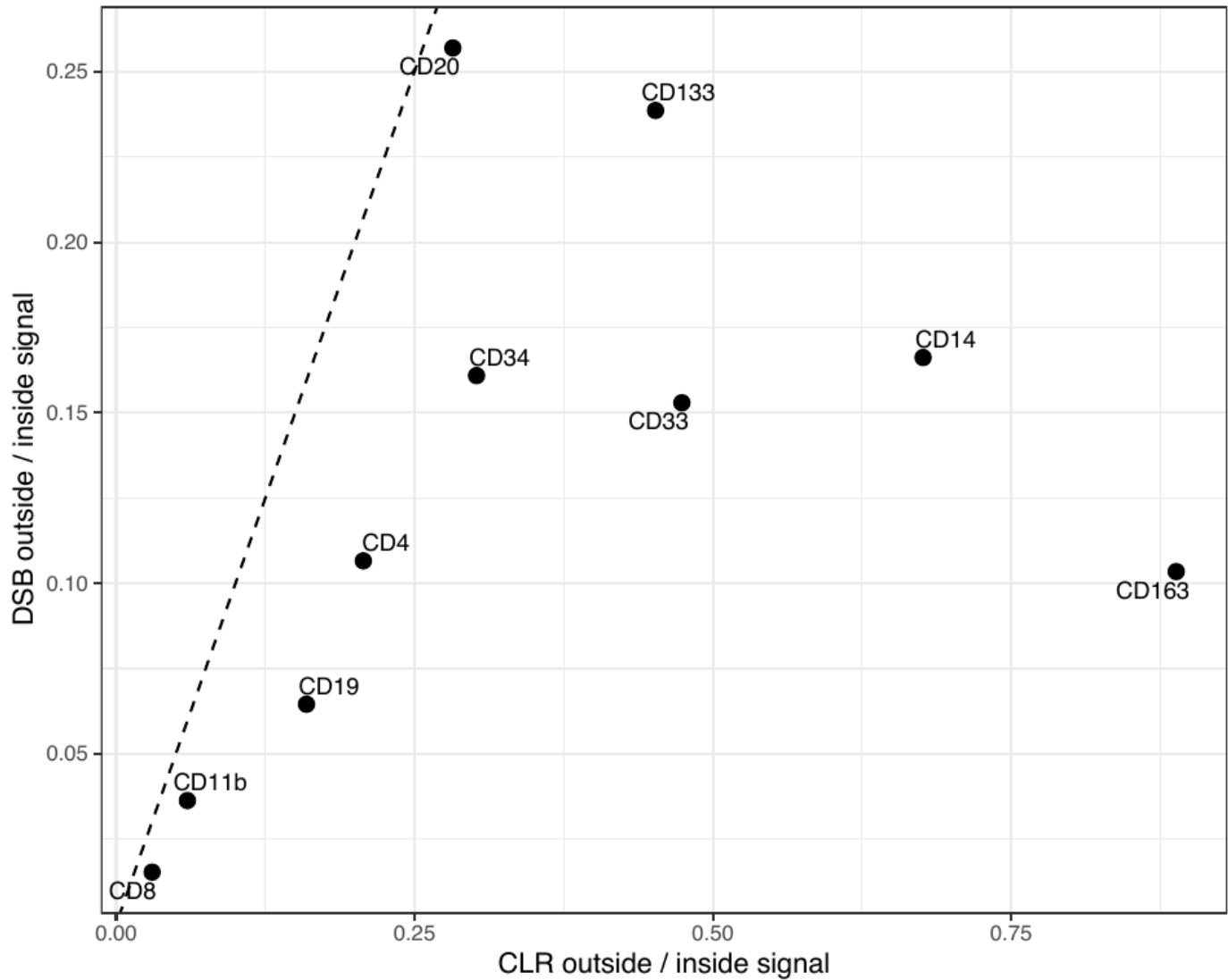
Marker signal outside the population it belongs to, on the same expected-marker panel. Per marker the script computes `inside_mean` (mean signal in the expected population), `outside_mean` (mean everywhere else), the percentage of outside cells scoring above zero, and

`leakage_ratio = outside_mean / inside_mean`. Lower is better: a genuinely specific marker should carry little signal outside its own population.

`03_marker_leakage_diffusion.csv` is sorted by `delta_leakage_ratio_DSB_minus_CLR`, so the markers DSB cleans up most are at the top.

Marker leakage outside expected population

Below diagonal means DSB reduced off-target/background spread



Background suppression

marker	mean_CLR	mean_DSB	delta_DSB_minus_CLR
IgM	2.8168731	-0.1249694	-2.9418425
IgG-Fc	2.6873402	-0.0377656	-2.7251058
Mouse-IgG2a	0.5459849	0.0542048	-0.4917802
Rat-IgG1-l	0.3367272	0.1759701	-0.1607571
Rat-IgG2c-k	0.2259942	0.0989940	-0.1270002
FCER1-alpha	0.1691688	0.0823210	-0.0868477
Mouse-IgG2b	0.1457389	0.1309483	-0.0147906
Armenian-Hamster-IgG	0.1067859	0.1315473	0.0247614
Rat-IgG2a-k	0.1058658	0.1784481	0.0725823
Mouse-IgG1	0.0900038	0.1976486	0.1076448
Rat-IgG2b	0.0376943	0.1582002	0.1205059
Rat-IgG1-k	0.0618827	0.1836524	0.1217696

Expected marker specificity

marker	expected_broad	expected_mean_CLR	expected_mean_DSB	other_median_CLR	other_median_DSB	specificity_CLR	specificity_DSB	expected_pct_p
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marker	expected_broad	expected_mean_CLR	expected_mean_DSB	other_median_CLR	other_median_DSB	specificity_CLR	specificity_DSB	expected_pct_p
CD8	Lymphoid	1.0951425	15.6111854	0.0000000	-0.1631998	1.0951425	15.7743852	
CD11b	Myeloid / DC	1.8591633	9.3511384	0.0000000	0.0931207	1.8591633	9.2580177	
CD4	Lymphoid	0.7939794	4.6337958	0.0000000	0.1094111	0.7939794	4.5243847	
CD19	Lymphoid	0.2405145	2.8677163	0.0000000	-0.0869771	0.2405145	2.9546933	
CD133	Stem / progenitor	2.1438014	3.7335483	0.8987586	0.5035712	1.2450427	3.2299771	
CD34	Stem / progenitor	4.2281987	4.5069696	0.9300531	0.1510431	3.2981456	4.3559264	
CD20	Lymphoid	0.0448804	0.9003657	0.0000000	0.0872981	0.0448804	0.8130676	
CD14	Myeloid / DC	0.7182740	0.8920683	0.4792751	0.1233388	0.2389989	0.7687294	
CD163	Myeloid / DC	1.6092160	0.5717646	1.4400683	0.0495966	0.1691477	0.5221680	
CD33	Myeloid / DC	1.3813027	1.1280148	0.6163499	0.0929426	0.7649528	1.0350722	

Marker leakage

marker	expected_broad	inside_mean_CLR	inside_mean_DSB	outside_mean_CLR	outside_mean_DSB	outside_positive_CLR	outside_positive_DSB	leakage_r
CD163	Myeloid / DC	1.6092160	0.5717646	1.4293611	0.0591313	99.839829	53.425877	0.8
CD14	Myeloid / DC	0.7182740	0.8920683	0.4856492	0.1482343	89.250756	56.807261	0.6
CD33	Myeloid / DC	1.3813027	1.1280148	0.6547175	0.1724860	89.980424	59.156433	0.4
CD133	Stem / progenitor	2.1438014	3.7335483	0.9688652	0.8908496	97.928994	71.745562	0.4
CD34	Stem / progenitor	4.2281987	4.5069696	1.2766819	0.7250150	96.745562	60.798817	0.3
CD4	Lymphoid	0.7939794	4.6337958	0.1644002	0.4937346	44.913987	60.284218	0.3
CD19	Lymphoid	0.2405145	2.8677163	0.0383500	0.1850282	13.219895	30.572177	0.3
CD20	Lymphoid	0.0448804	0.9003657	0.0126601	0.2313271	5.422588	64.285714	0.3
CD11b	Myeloid / DC	1.8591633	9.3511384	0.1108525	0.3393169	27.015483	62.270867	0.6
CD8	Lymphoid	1.0951425	15.6111854	0.0329397	0.2389299	8.937921	8.937921	0.6

5.6 CNV calls (CopyKAT)

The next challenge was to identify cancer vs normal cells within individual samples. CopyKAT was run twice: once on all cells pooled (step 16) and then per donor (step 17). The per-donor run produced better results and is what produced the output below.

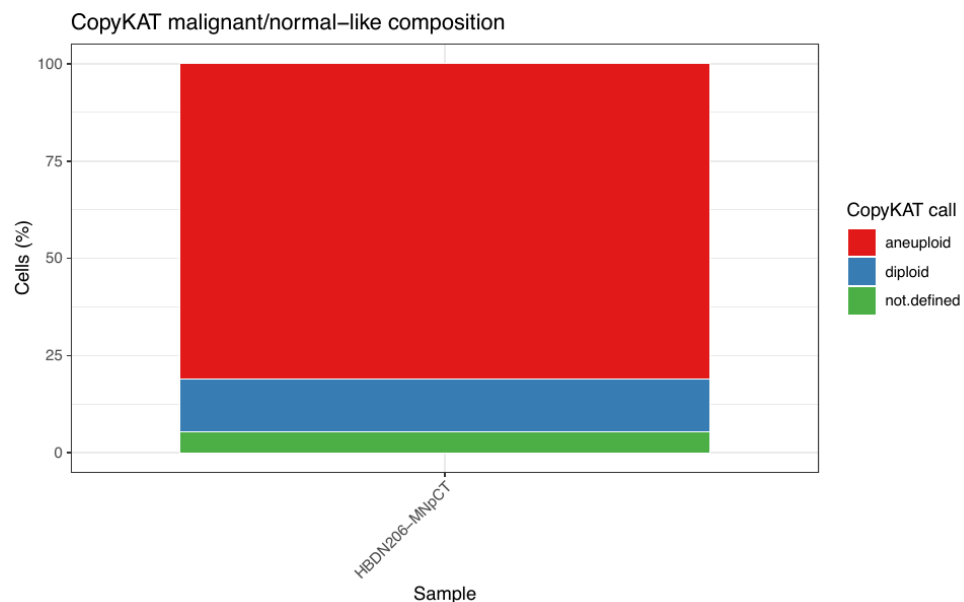
HBDN206-MNpCT

CopyKAT calls - HBDN206-MNpCT

sampleID	copykat_call	n_cells	percent
HBDN206-MNpCT	aneuploid	4619	81.077760
HBDN206-MNpCT	diploid	774	13.586098
HBDN206-MNpCT	not.defined	304	5.336142

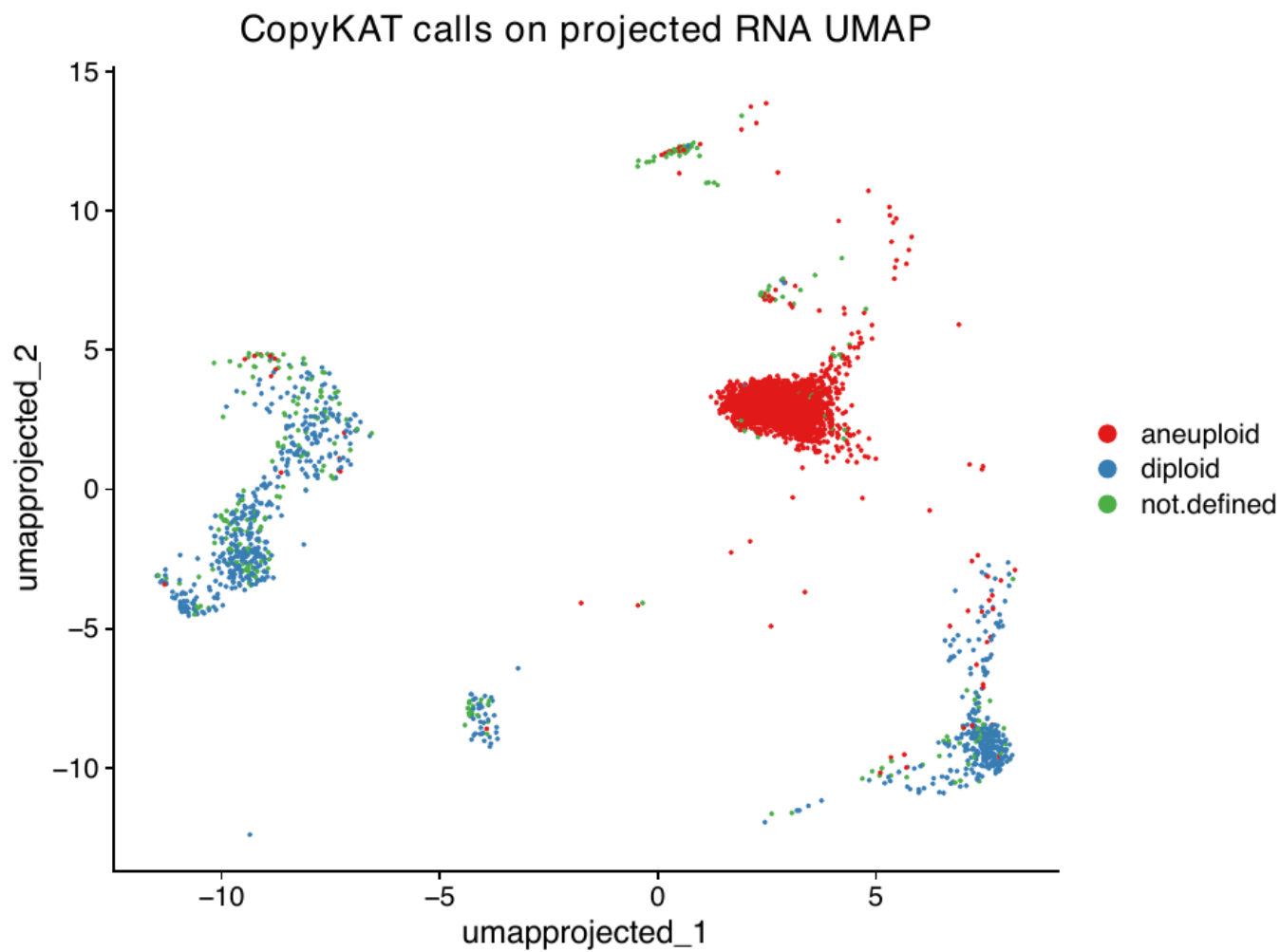
01_copykat_call_composition_by_sampleID.pdf

Aneuploid / diploid proportions.



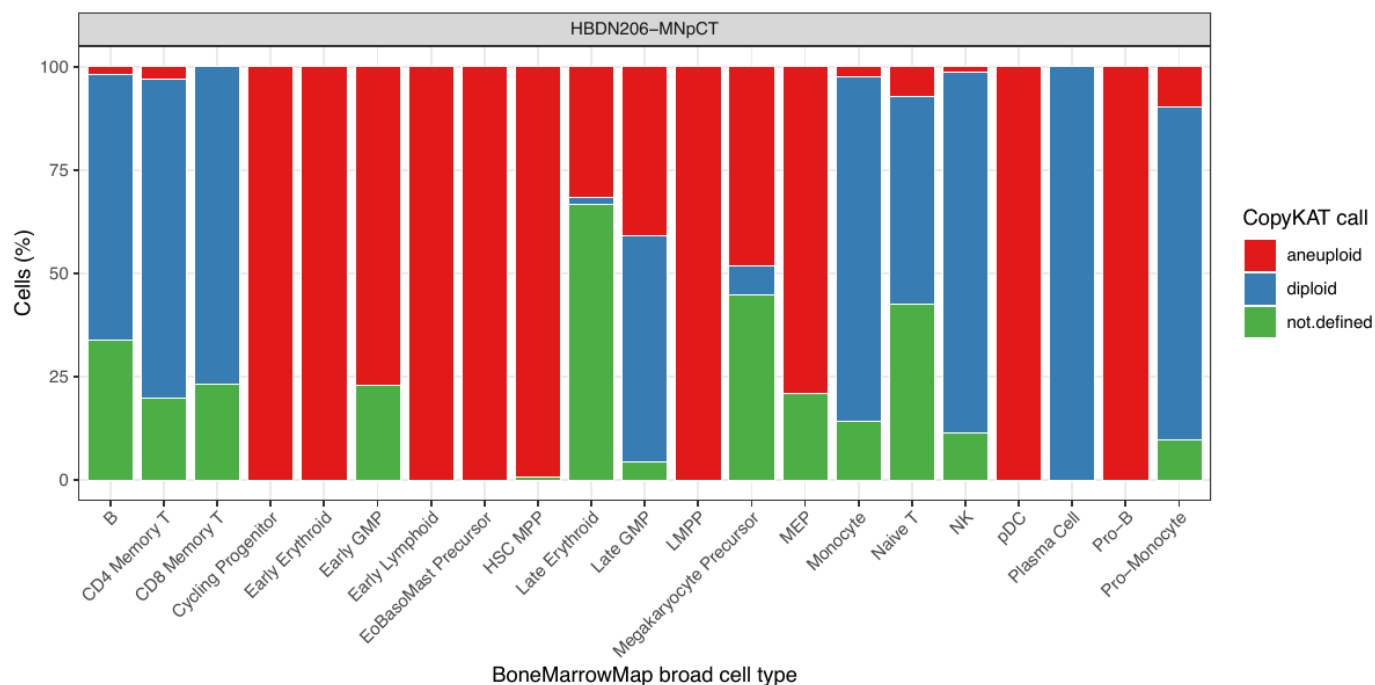
02_copykat_calls_on_projected_RNA_UMAP.pdf

CopyKAT calls on the projected UMAP.



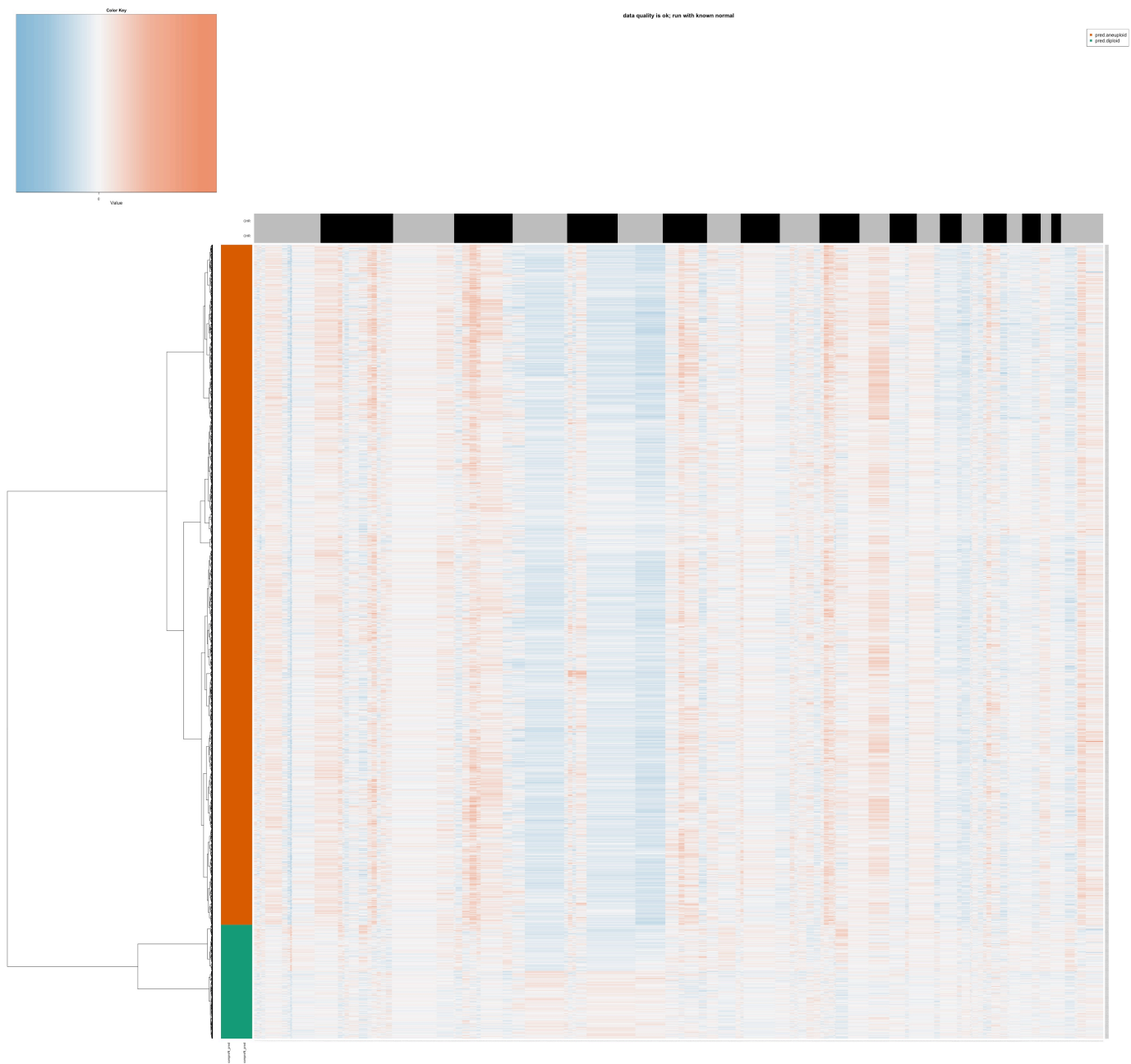
03_copykat_by_BoneMarrowMap_broad_celltype.pdf

Calls against annotation. The check that 'aneuploid' is not just landing on one cell type for technical reasons.



LK1_copykat_copykat_heatmap.jpeg

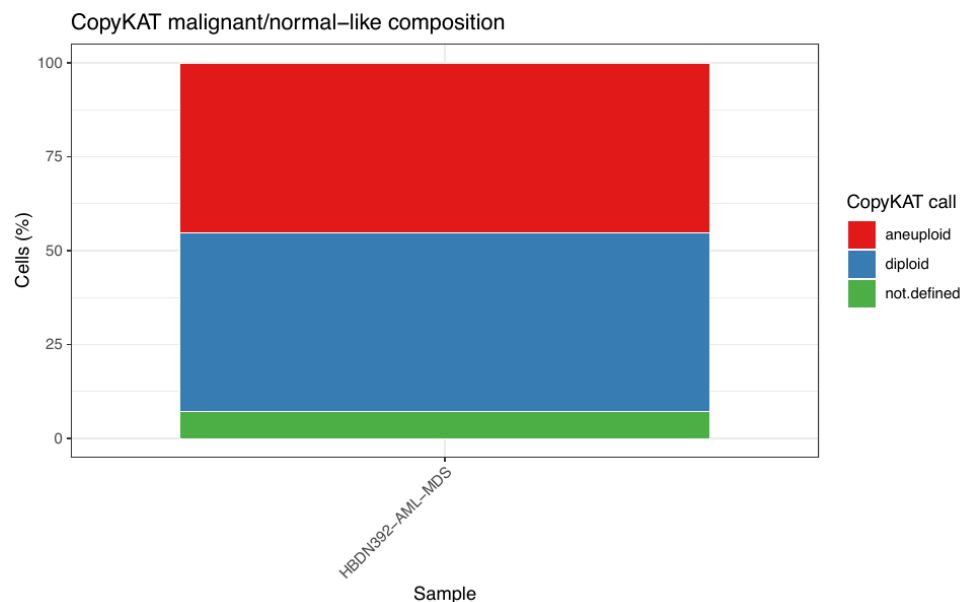
The CopyKAT CNV heatmap.



HBDN392-AML-MDS

01_copykat_call_composition_by_sampleID.pdf

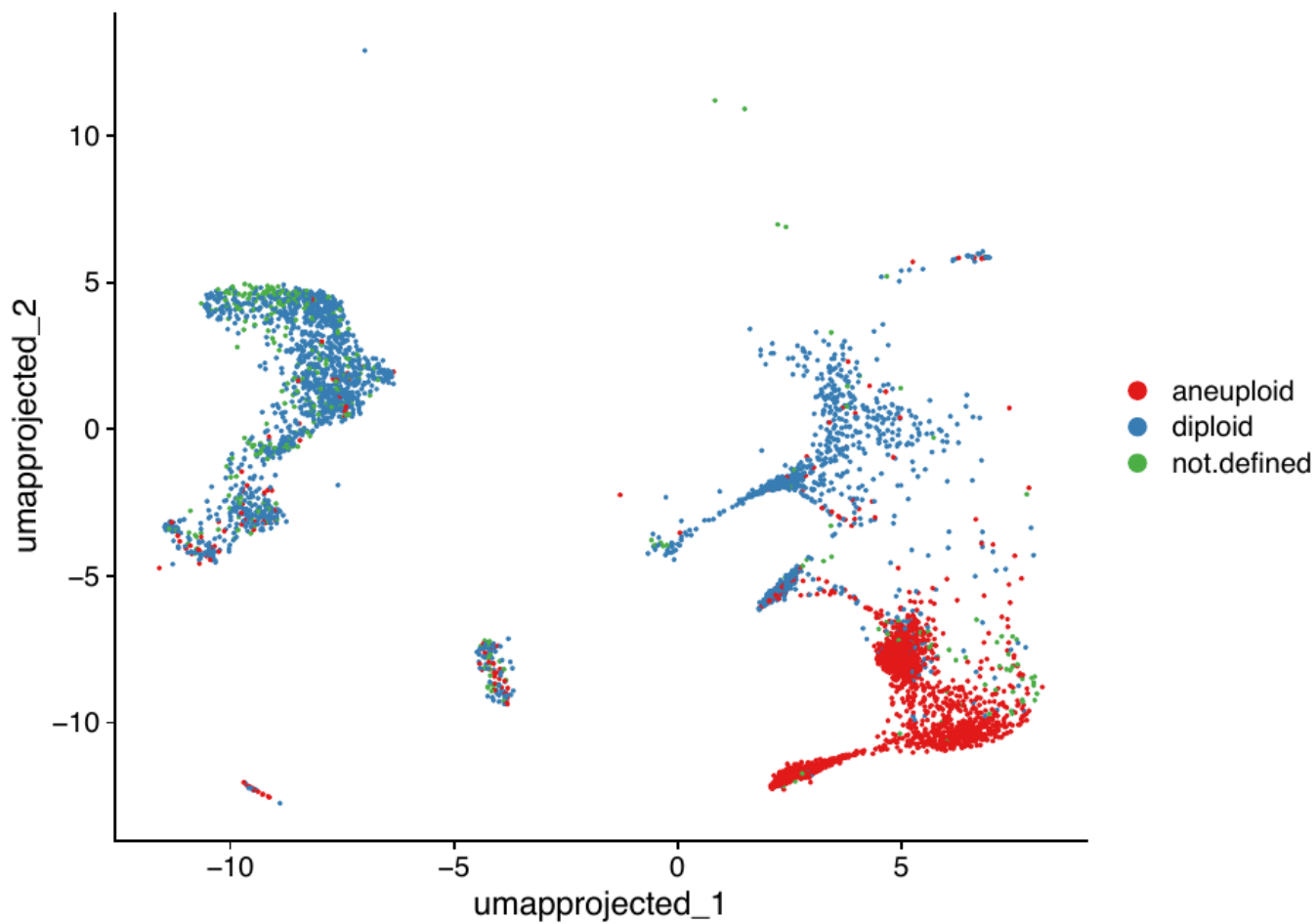
Aneuploid / diploid proportions.



[02_copykat_calls_on_projected_RNA_UMAP.pdf](#)

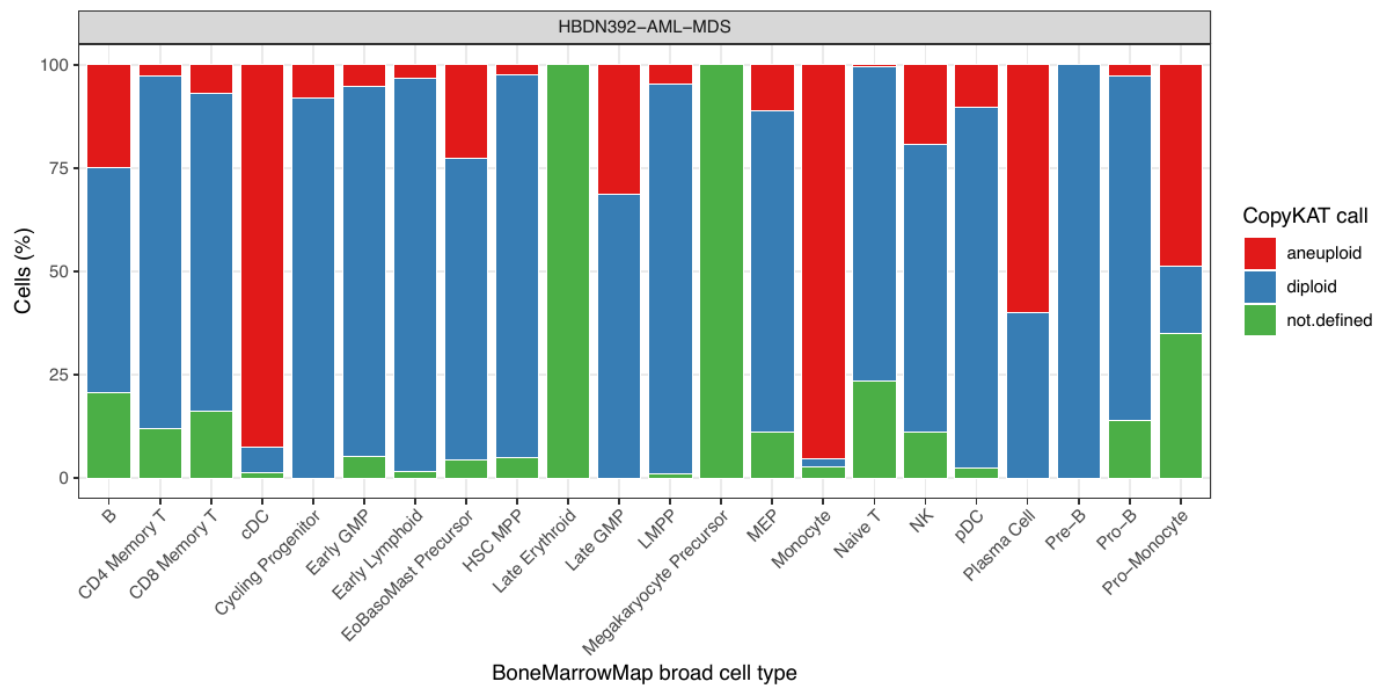
CopyKAT calls on the projected UMAP.

CopyKAT calls on projected RNA UMAP



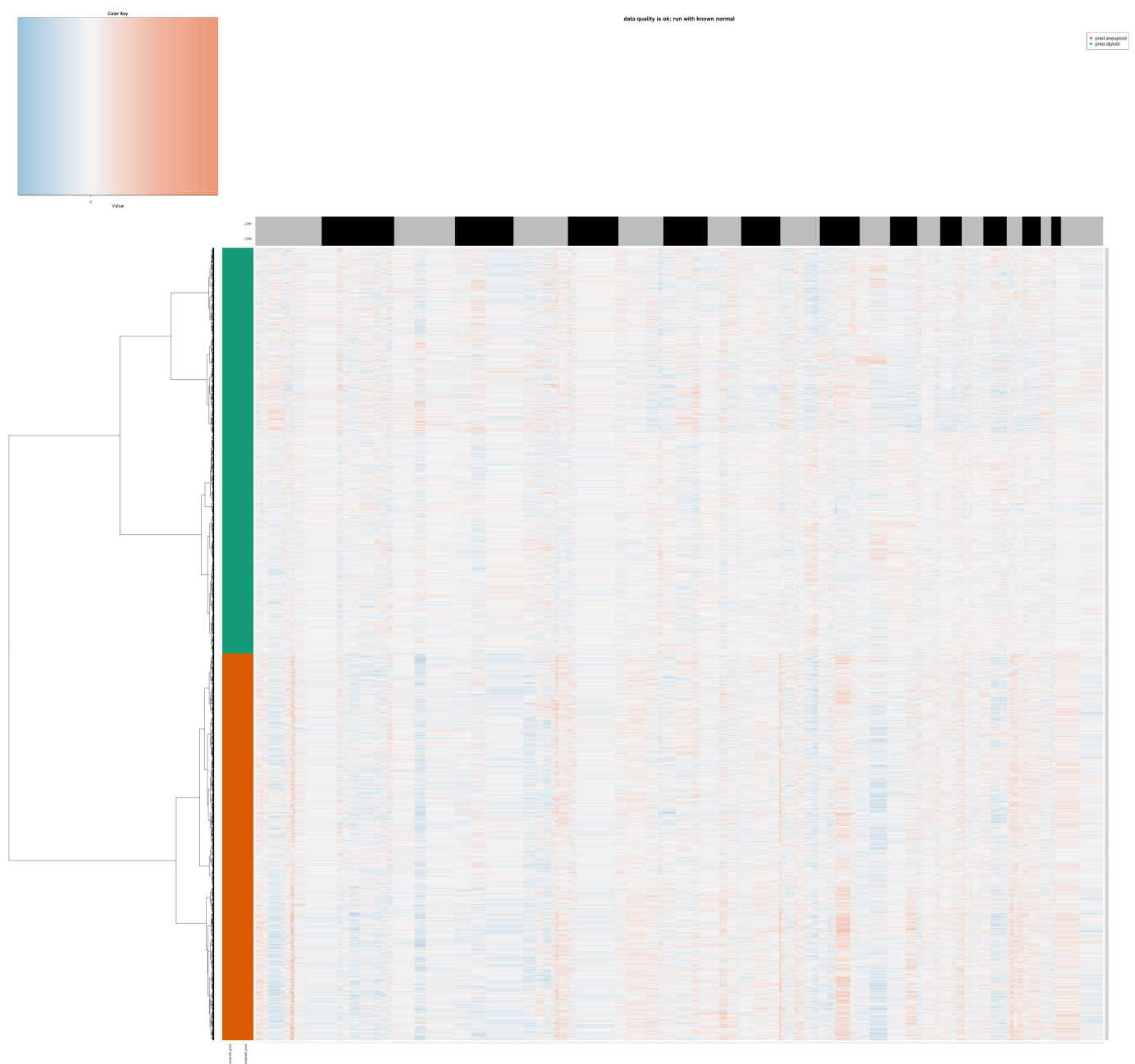
[03_copykat_by_BoneMarrowMap_broad_celltype.pdf](#)

Calls against annotation. The check that 'aneuploid' is not just landing on one cell type for technical reasons.



LK1_copykat_copykat_heatmap.jpeg

The CopyKAT CNV heatmap.



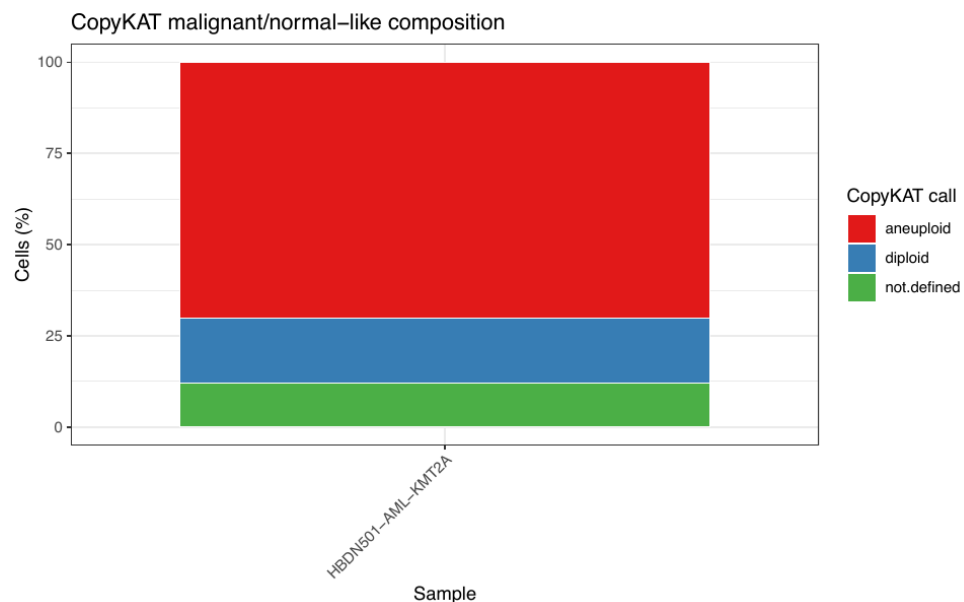
HBDN501-AML-KMT2A

CopyKAT calls - HBDN501-AML-KMT2A

sampleID	copykat_call	n_cells	percent
HBDN501-AML-KMT2A	aneuploid	1965	70.10346
HBDN501-AML-KMT2A	diploid	498	17.76668
HBDN501-AML-KMT2A	not.defined	340	12.12986

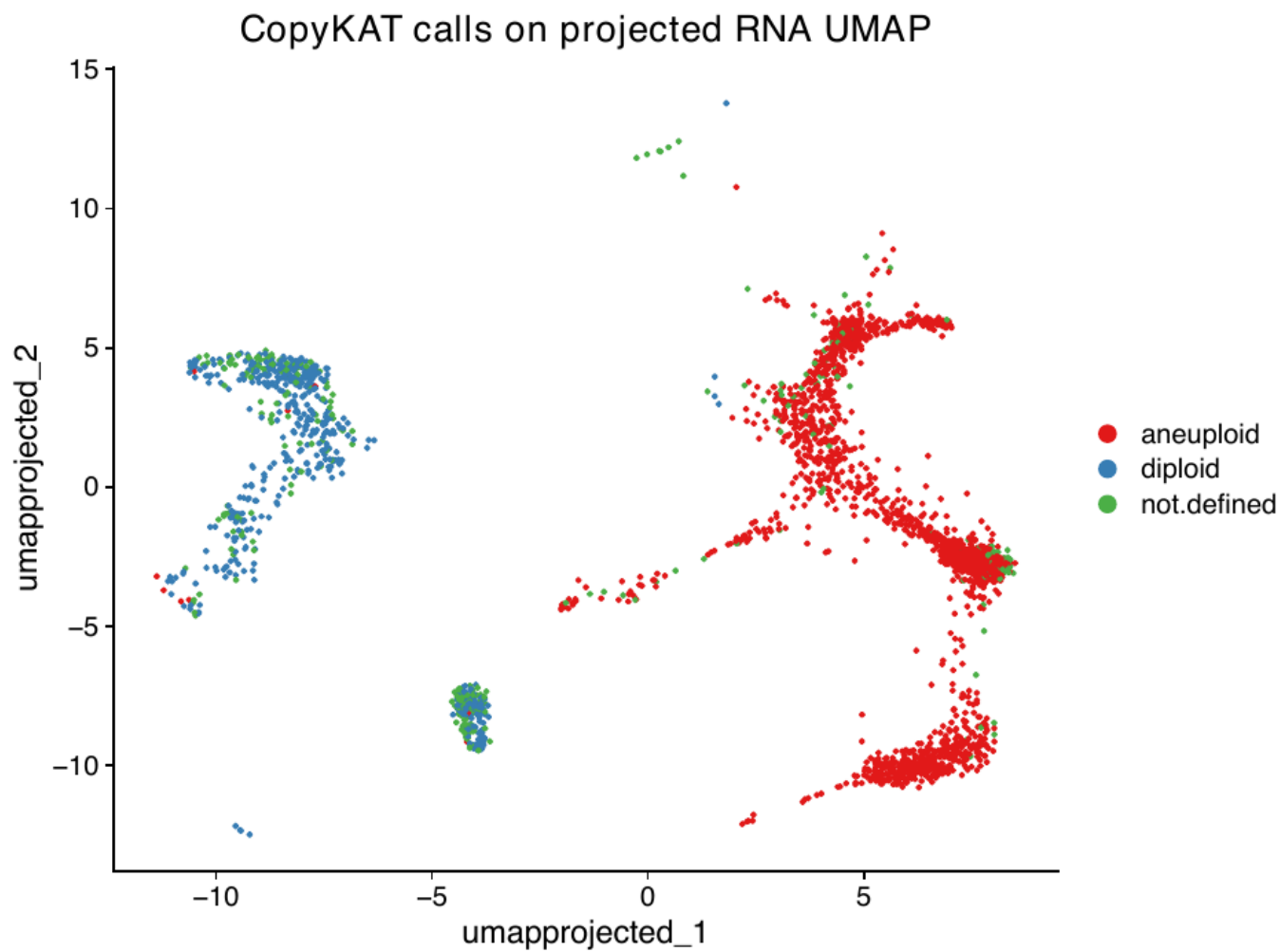
01_copykat_call_composition_by_sampleID.pdf

Aneuploid / diploid proportions.



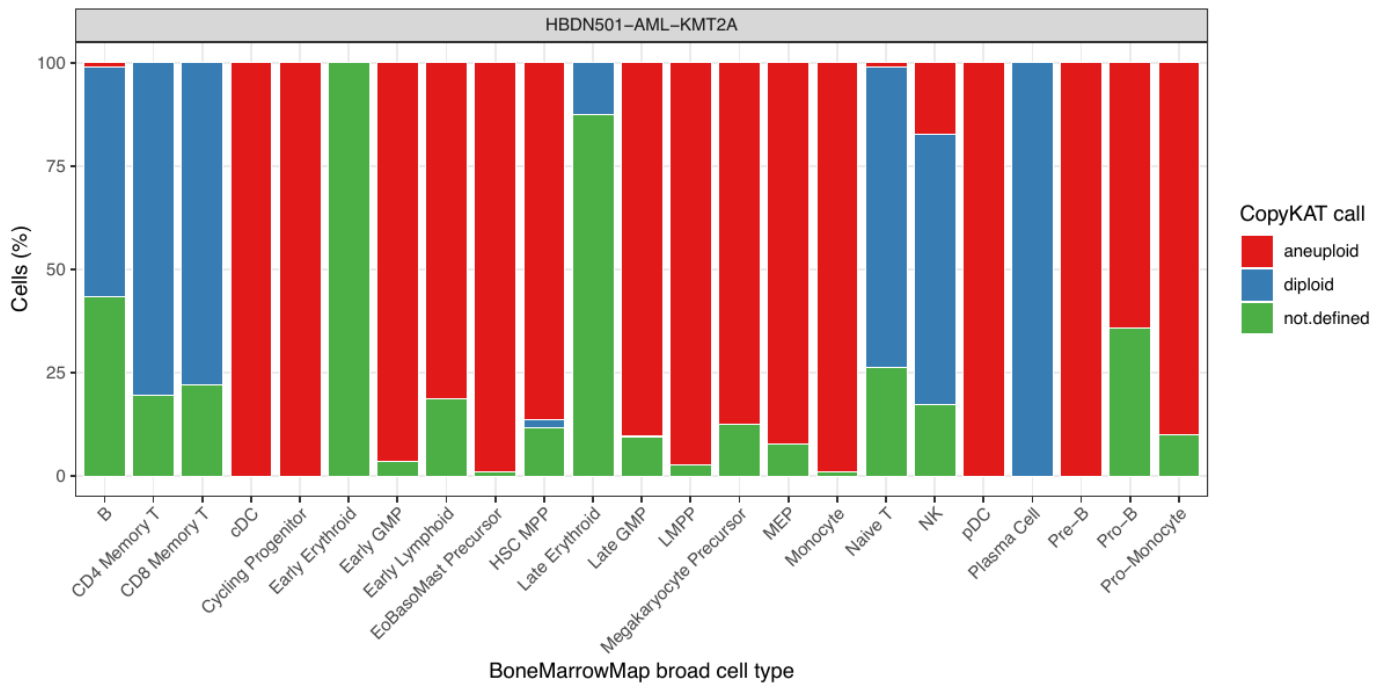
02_copykat_calls_on_projected_RNA_UMAP.pdf

CopyKAT calls on the projected UMAP.



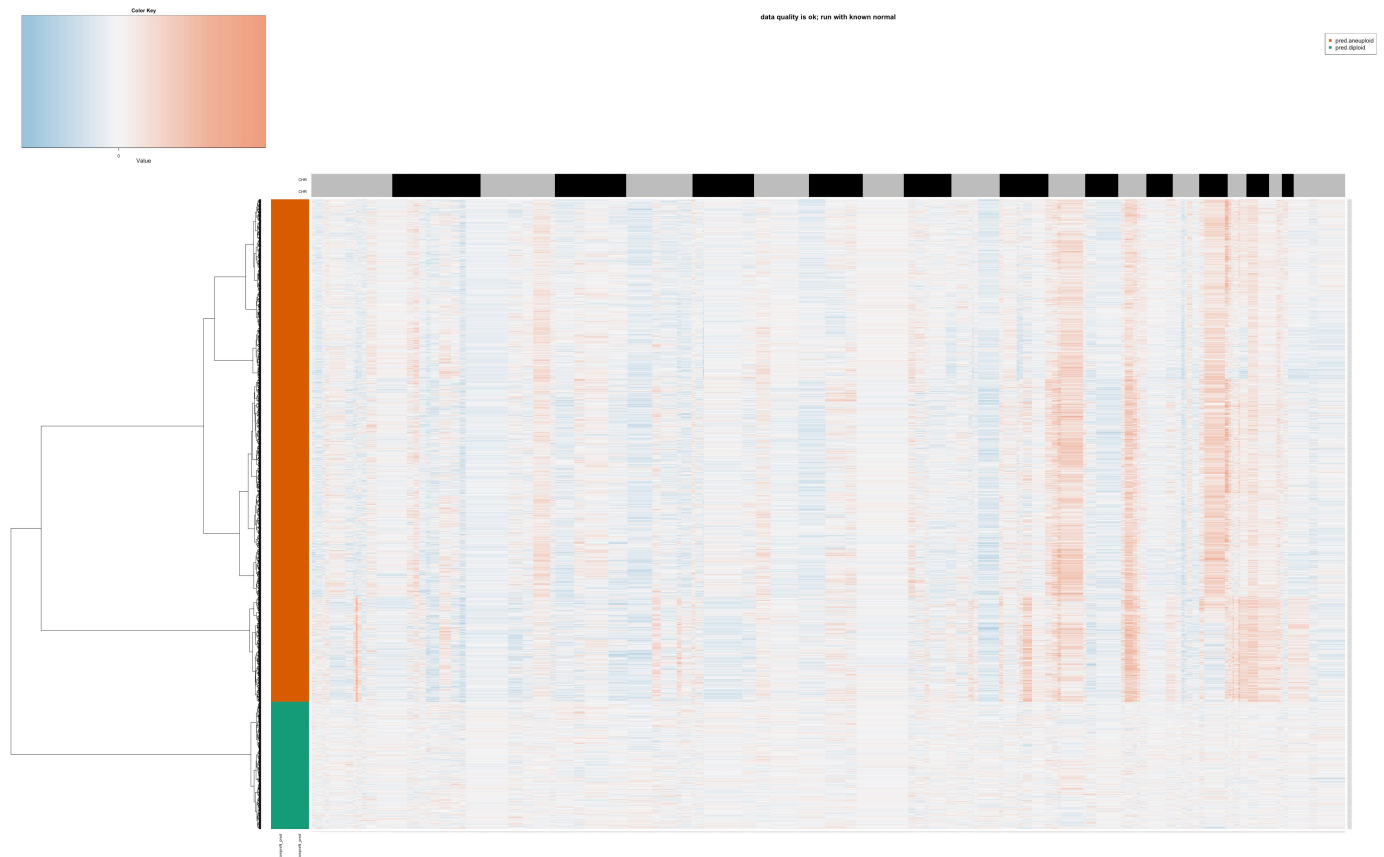
03_copykat_by_BoneMarrowMap_broad_celltype.pdf

Calls against annotation. The check that 'aneuploid' is not just landing on one cell type for technical reasons.



LK1_copykat_copykat_heatmap.jpeg

The CopyKAT CNV heatmap.



6. Tools

Tool	Version	Source
Cell Ranger	10.0.0	~/tools/cellranger-10.0.0/cellranger
Cell Ranger annotation	refdata-gex-GRCh38-2024-A	annotation/
CITE-seq-Count	to confirm	scripts/3.hto_count.sbatch , 4.cite_seq_count.sbatch

Tool	Version	Source
cellsnp-lite	1.2.3 (htslib 1.23.1)	cellsnp_env ,used by 04.cellsNP_vireo.sh
vireo (vireoSNP)	to confirm	cellsnp_env ,used by 04.cellsNP_vireo.sh

R side, the session that renders this report:

```
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5.2
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] scales_1.4.0  ggplot2_4.0.2 knitr_1.51    purrr_1.2.1  stringr_1.6.0
## [6] readr_2.1.6   tidyr_1.3.2   dplyr_1.2.0
##
## loaded via a namespace (and not attached):
## [1] bit_4.6.0      gtable_0.3.6   jsonlite_2.0.0  crayon_1.5.3
## [5] qpdf_1.4.1     compiler_4.5.1 pdftools_3.9.0  Rcpp_1.1.1
## [9] tidyselect_1.2.1 parallel_4.5.1  jquerylib_0.1.4  yaml_2.3.12
## [13] fastmap_1.2.0  R6_2.6.1       labeling_0.4.3   generics_0.1.4
## [17] tibble_3.3.1   RColorBrewer_1.1-3 bslib_0.10.0    pillar_1.11.1
## [21] tzdb_0.5.0     rlang_1.1.7    cachem_1.1.0    stringi_1.8.7
## [25] xfun_0.56      S7_0.2.1       sass_0.4.10     bit64_4.6.0-1
## [29] otel_0.2.0     cli_3.6.5      withr_3.0.2     magrittr_2.0.4
## [33] digest_0.6.39  grid_4.5.1     vroom_1.7.0     rstudioapi_0.18.0
## [37] askpass_1.2.1  hms_1.1.4      lifecycle_1.0.5 vctrs_0.7.1
## [41] evaluate_1.0.5 glue_1.8.0     farver_2.1.2    rmarkdown_2.30
## [45] tools_4.5.1    pkgconfig_2.0.3 htmltools_0.5.9
```

7. Caveats

1. **The run is superseded.** LK1 was re-sequenced as 260522_VH01624_461_222JLVNX , and the re-sequenced data is what later analyses use.

Scripts dropped when collating this folder, and why:

Dropped	Reason
backup/8.create_seurat_object.R	Byte-identical to scripts/6a.initial_QC.R (step 03).
backup/10b.annotate_DSB.R	Byte-identical to scripts/12.identify_cancer_cells.R (step 12).
backup/4.initial_QC.R	Earlier, shorter version of step 03.
backup/11a.example_cite_copykat.R	Exploratory scratch, not part of the pipeline.
add_cite_for_hiep.R	One-off to hand CITE data to a colleague, no analytical result.

Note that scripts/12.identify_cancer_cells.R does not identify cancer cells - it is the BoneMarrowMap projection on the DSB object, byte-identical to backup/10b.annotate_DSB.R. The name is misleading, so it is 12.annotate_CITE_DSB.R here.